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Vinod et al.

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(45) **Date of Patent: Aug. 26, 2025**

(54) **ARTIFICIAL INTELLIGENCE BASED
TRANSACTIONS CONTEXTUALIZATION
PLATFORM**

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K Praveen**, Bengaluru (IN)

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U.S.C. 154(b) by 0 days.

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(21) Appl. No.: **18/387,799**

(57)

ABSTRACT

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US 2025/0148515 A1 May 8, 2025

(51) **Int. Cl.**

G06Q 30/0601 (2023.01)

G06N 7/02 (2006.01)

G06N 20/00 (2019.01)

(52) **U.S. Cl.**

CPC **G06Q 30/0623** (2013.01); **G06N 7/02**
(2013.01); **G06N 20/00** (2019.01)

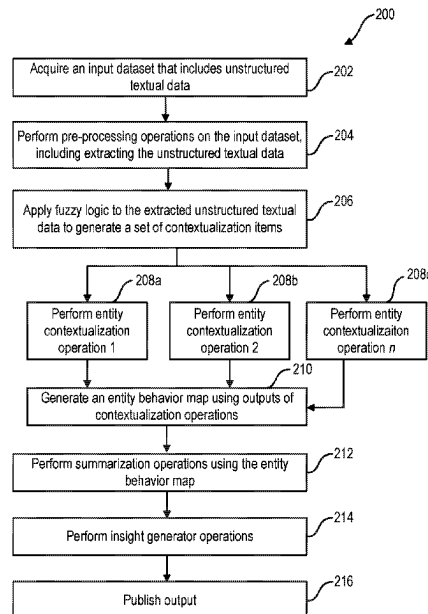
(58) **Field of Classification Search**

CPC G06N 7/02; G06N 20/00; G06N 5/048;
G06N 5/045; G06Q 30/0623; G06Q
10/0875; G06F 16/215; G06F 16/2468

A data analytics and contextualization platform instance acquires an input dataset. Using fuzzy logic, the platform generates a set of contextual items based on the unstructured textual data. Items in the input dataset are labeled using contextual items. Using a labeled particular item, a set of entity behavior maps can be generated to include one or more features selected based on a first particular label of the particular item. One or more AI/ML modeling operations can be selected based on a type of the entity behavior map and performed for items in the entity behavior map. A particular first subset of AI/ML modeling operations can be performed, using a particular contextual item as a trigger, in parallel relative to a particular second subset of AI/ML modeling operations. The entity behavior map, which can be populated using outputs of the AI/ML operations, can be transmitted to a target computing system.

See application file for complete search history.

21 Claims, 19 Drawing Sheets



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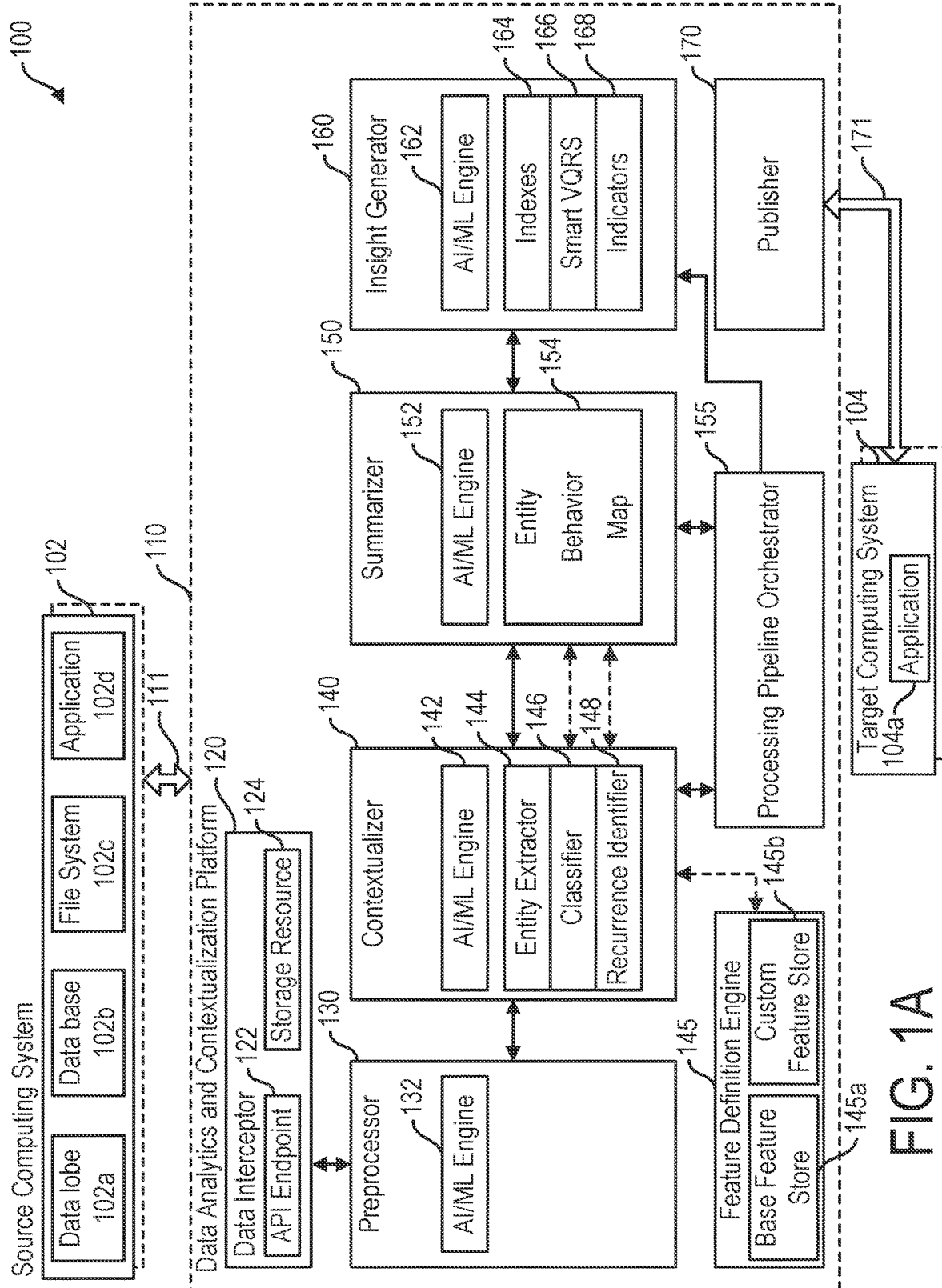


FIG. 1A

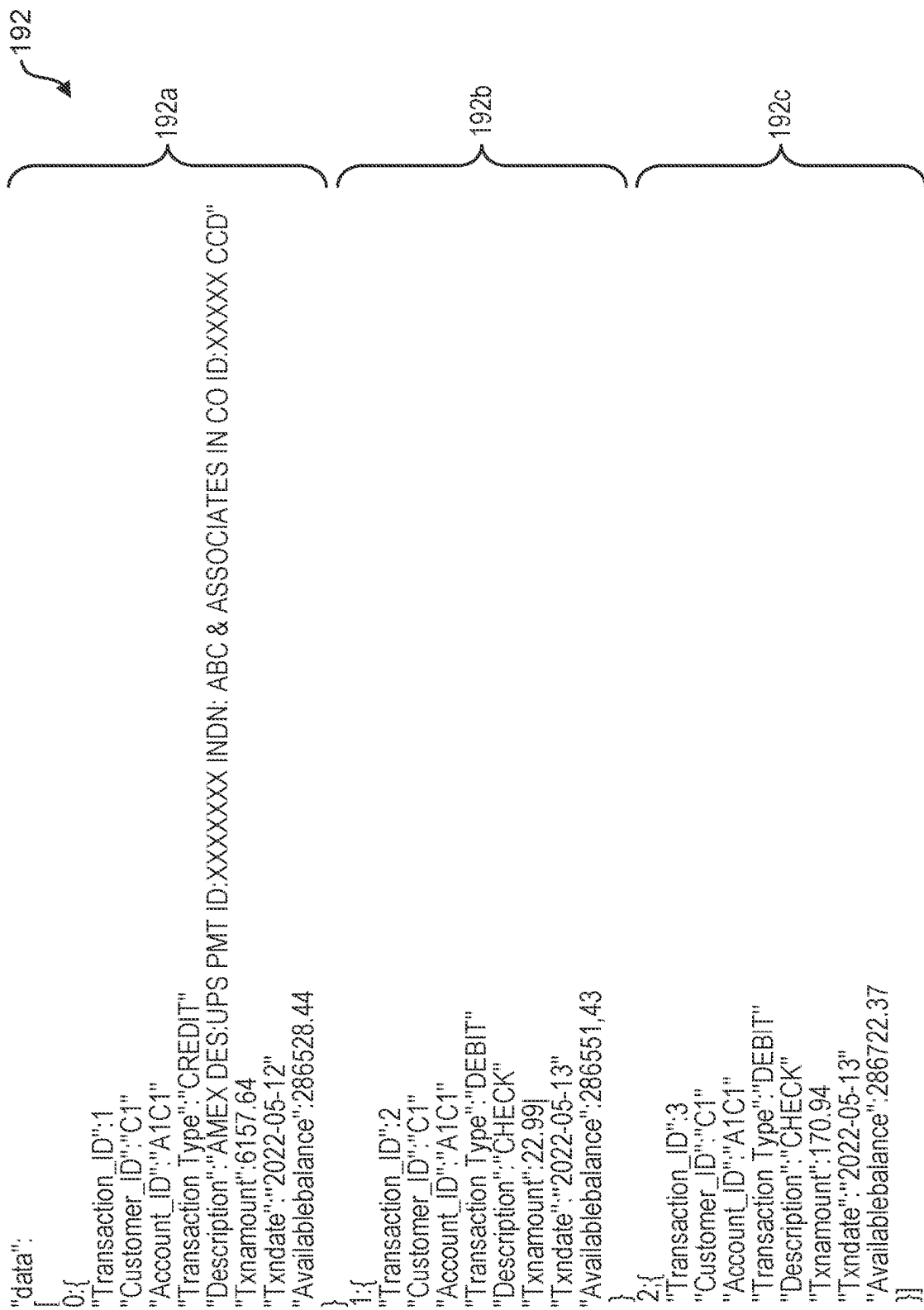


FIG. 1B

194 { {
"Customer_ID": "C1",
"Transaction_ID": 1,
"Account_ID": "A1C1",
"TransactionType": "CREDIT",
"Description": "AMEX DES:UPS PMT ID:XXXXXXX INDN:ABC & ASSOCIATES IN CCD",
"Txnamount": 6157.64,
"Txndate": "2022-05-12 00:00:00",
"TransactionCategoryD1": "CHARGE VOLUME",
"TransactionCategoryD2": "PROCESSOR CV",
"TransactionCategoryD3": "PAYMENT NETWORK",
"VendorName": null,
"InstitutionName": "AMERICAN EXPRESS",
"RecurrenceType": 3,
"RecurrenceFrequency": "daily",
"IndAutoPay": 0, }
"Channel": "Ach/Electronic Transfer",
"City": null, "State": null }]

FIG. 1C

```
{ "Customer_ID": "C1",  
  "Tradeline_ID": "xxx1",  
  "Tradeline_Type": "Financial",  
  "Tradeline_subtype": "BUSINESS LOAN PAYMENT",  
  "InstitutionName": "ASCENTIUUM",  
  "LenderType": "Non Traditional Lender",  
  "RecurrenceFrequency": "other",  
  "RecurrenceType": 3,  
  "Payment_M1": 378.2,  
  "Payment_M2": 378.2,  
  "Payment_M3": null,  
  "Payment_M4": 673.2,  
  "Payment_M5": null,  
  "Payment_M6": null,  
  "PaymentTicketSize": 378.2,  
  "LastPaymentDate": "2022-04-17 00:00:00",  
  "LastPaymentAmount": 378.2,  
  "ShareofPayment": 1.0    },  
{ "Customer_ID": "C1",  
  "Tradeline_ID": "xxx2",  
  "Tradeline_Type": "Financial",  
  "Tradeline_subtype": "CREDIT CARD PAYMENT",  
  "InstitutionName": "AMERICAN Express",  
  "LenderType": "Traditional Lemder",  
  "RecurrenceFrequency": "None",  
  "RecurrenceType": 0,  
  "Payment_M1": null,  
  "Payment_M2": null,  
  "Payment_M3": null,  
  "Payment_M4": 660.33,  
  "Payment_M5": null,  
  "Payment_M6": 515.09,  
  "PaymentTicketSize" : 428.56,  
  "LastPaymentDate": " 2022-01-16 00:00:00",  
  "LastPaymentAmount": 231.77,  
  "Shareofpaymet": 0.118  }
```

196a

196b

FIG. 1D

```
[ {
  "Customer_ID": "C1",
  "AutoLoanIndicator": 1,
  "BusinessLoanIndicator": 1,
  "CashInHandDays": 31,
  "CoefVarCreditCount": 0.16,
  "CoefVarDeditCount": 0.11,
  "CoefVarRevenue": 0.17,
  "CommercialLoyaltyIndex": 1,
  "CommercialRiskIndex": 3,
  "DebtRepaymentMonth1": 47255,
  "DollarATMCredit": 10220,
  "DSCR1": 0.1,
  "Electricity/Gas/PowerPaymentCount": 4,
  "GrossProfitL1byP1": 0.65,
  "InsuranceIndicator": 0,
  "MOMTotalDebtRepayment1": 0.727,
  "MortgageIndicator": 0,
  "NSFCountTotal": 2,
  "NSFLastDays": 37,
  "NSFTotalAmount": 70,
  "NumBranchCredit": 21,
  "NumDigitalWalletDebitMonth1": 0,
  "NumLoanCredits": 9,
  "NumLoanCreditsMonth1": 1,
  "OverdraftLimit": 0,
  "PayrollFrequency": others,
} ]
```



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FIG. 1E

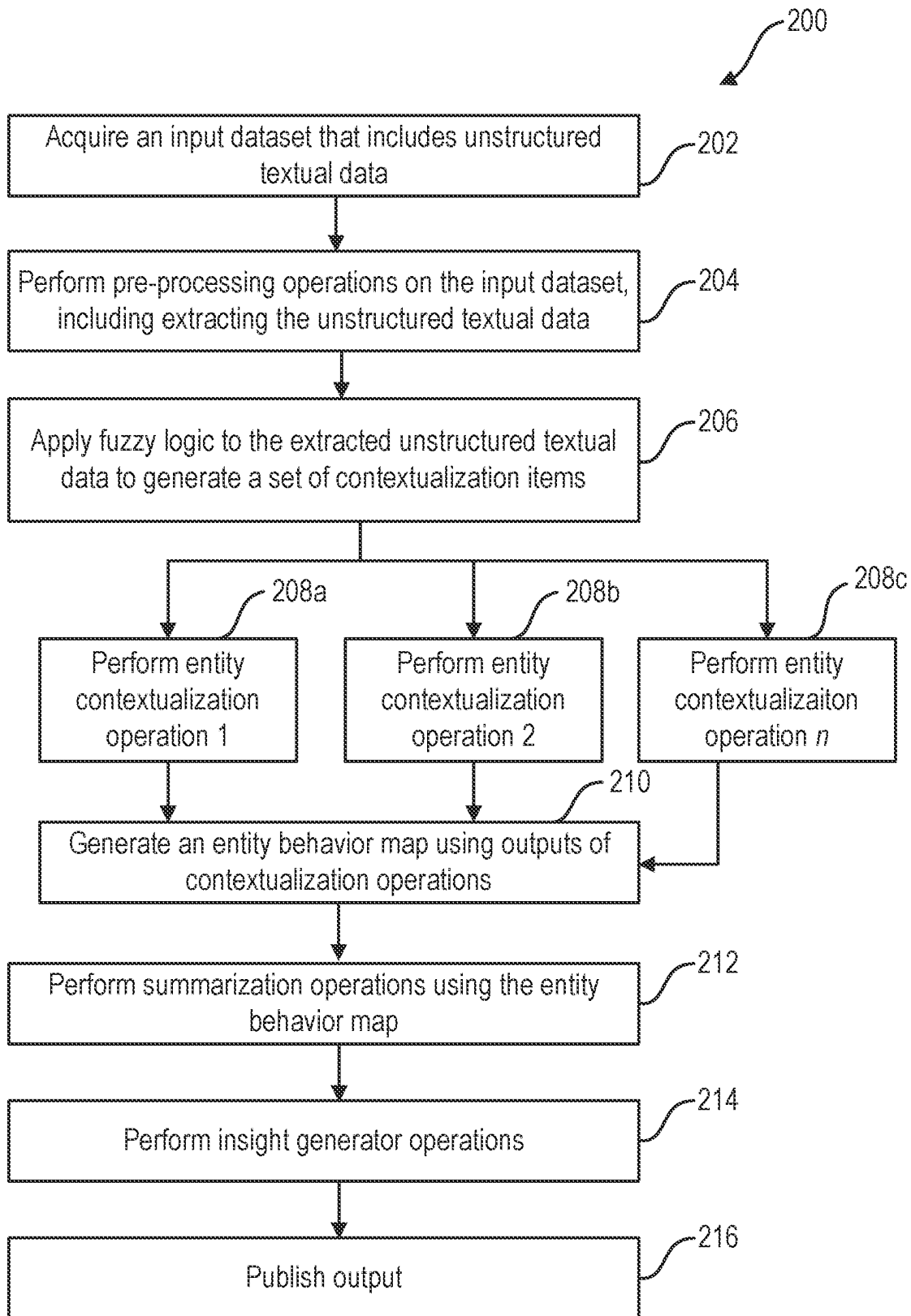


FIG. 2A

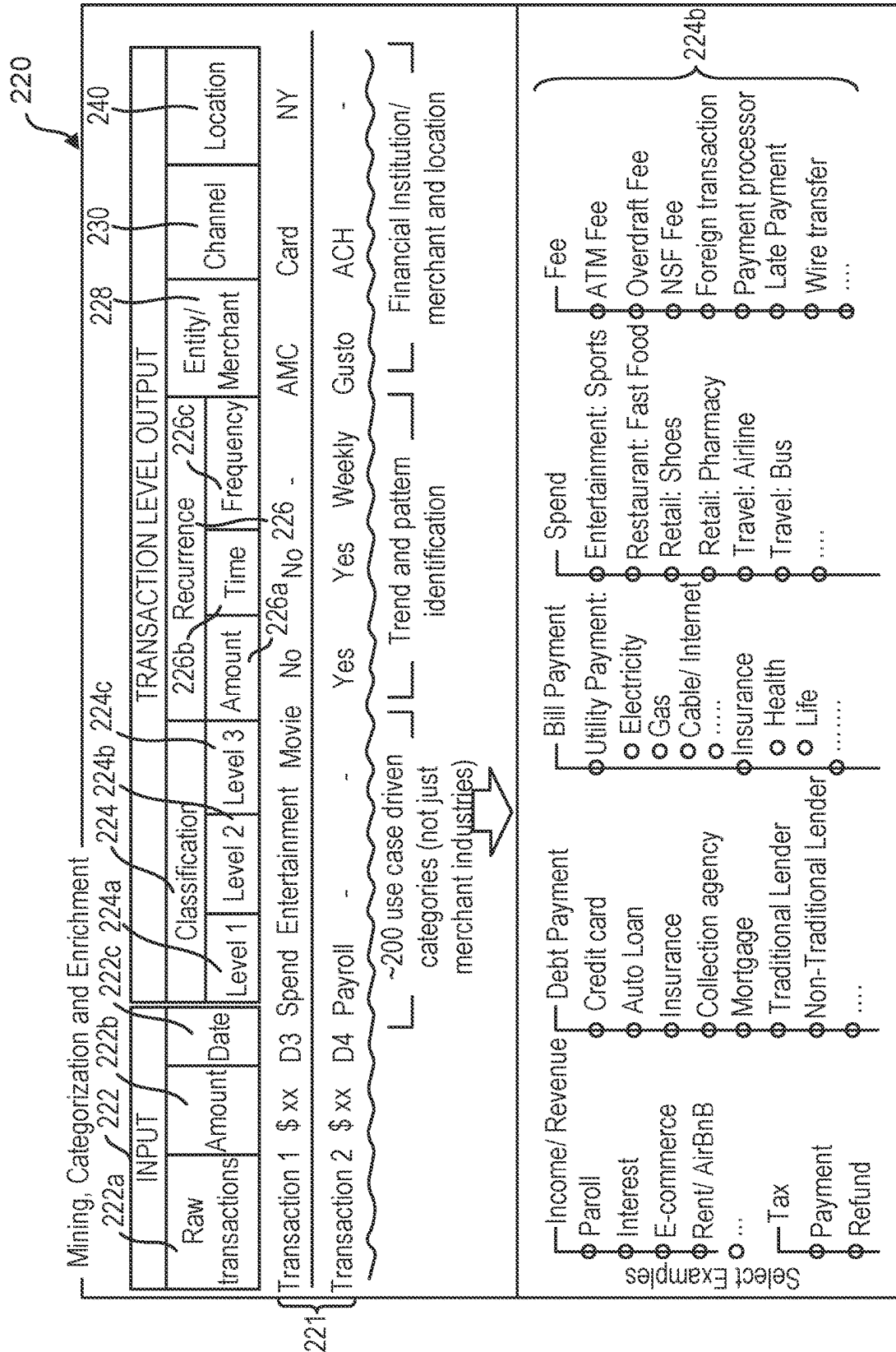


FIG. 2B

252 Description	254 Amount	256 Date	258 Category: Institution		260 Category: Level 2		262 Category: Level 3		264 Recurrence Type	Frequency	250
			Level 1	Name	Level 2		Level 3				
AMZN IQ2HYUSL MARKETPLAC xxHVC7H CCD ID: xx246	\$503	1-Jul	Charge	VOLUME	AMAZON	Non processor CV	E-commerce	NA	NA	NA	* Revenue sources and stability
AMEX UPS PMT xx295 CCD ID: XX49	\$890	1-Jul	Charge	VOLUME	AMERICAN EXPRESS	Processor CV	Payment Network	NA	NA	NA	
AMERICAN EXPRESS ACH PMT W4320 WEB ID: xx11	(\$6,000)	5-Jul	Debt	obligation	AMERICAN EXPRESS	Credit Card Payment	Credit Card Payment	NA	NA	NA	* Payments towards financial tradelines and relationships
07/03 PAYMENT TO CHASE CARD ENDING IN 1573	(\$10,000)	5-Jul	Debt	obligation	JP MORGAN CHASE	Credit Card Payment	Credit Card Payment	NA	NA	NA	* Share of wallet and need
GEXA ENERGY ELECPAYMINT NONE PPD ID: xx923	(\$1,144)	2-Aug	Bill	Payment	GEXA ENERGY	Utilities	Electricity/ Gas/ Power	NA	NA	NA	* Consistent' electricity payment
KABBAGE INC LOAN 50974 PPD ID: xxxxx	\$75,000	28-May	Loan	Funding	KABBAGE	Business Loan Funding	Non-Traditional Lender	NA	NA	NA	* New business loan deposit
KABBAGE INC PAYMENT 50974 CCD ID: xxx761	(\$11,007)	15-Jul	Debt	obligation	KABBAGE	Business Loan Payment	Non-Traditional Lender	Desc & Freq	Monthly		* Recurrence (monthly) payment towards business loan (Kabbage)
KABBAGE INC PAYMENT 50974 CCD ID: xxx761	(\$11,007)	15-Aug	Debt	obligation	KABBAGE	Business Loan Payment	Non-Traditional Lender	Desc & Freq	Monthly		
KABBAGE INC PAYMENT 50974 CCD ID: xxx761	(\$14,082)	14-Sep	Debt	obligation	KABBAGE	Business Loan Payment	Non-Traditional Lender	Desc & Freq	Monthly		* Payment amount increase from \$11K to \$14K
ROCKET RECEIVABLES ONLINE PAYMENT xx21 WEB ID: xx	(\$445)	2-Oct	Debt	Payment	ROCKET RECEIVABLES	Collection Agency	Collection Agency	NA	NA	NA	* Payment to collection agency

FIG. 2C

Tradeline Subtype	Institution	Recurrence Type	Recurrence Frequency	284						Last Payment Amount	Avg. payment transaction size	Share of Payment
				Payment M1	Payment M2	Payment M3	Payment M4	Payment M5	Payment M6			
AUTO LOAN PAYMENT	HONDA FINANCE	Both Amount & Time	Monthly	\$1,125			\$375	\$375	\$375	\$1,125	\$375	100%
CREDIT CARD PAYMENT	JP MORGAN CHASE	None	None				\$39,862	\$27,876	\$62,698	\$5,124	\$3,707	15%
	JP MORGAN CHASE	None	None	\$44,882	\$41,055	\$48,071	\$25,840	\$18,550	\$22,431	\$10,000	\$4,427	24%
	AMERICAN EXPRESS	None	None	\$101,603	\$58,217	\$51,499	\$114,174	\$80,173	\$68,996	\$8,000	\$4,000	56%
	CAPITAL ONE	None	None	\$7,500	\$8,640		\$15,500	\$3,546		\$7,500	\$5,023	4%
	CITIBANK	None	None	\$7,090			\$3,104	\$95	\$1,090	\$2,097		1%
BUSINESS LOAN PAYMENT	KABBAGE	Only Time	Monthly	\$20,732	\$17,686	\$18,097	\$14,082	\$11,007	\$11,007	\$20,732	\$15,884	39%
	ONDECK	None	None	\$24,100	\$19,914	\$36,626	\$19,620	\$23,980	\$21,800	\$1,090	\$1,090	61%
	SHOPIFY	Both Amount & Time	Weekly	\$225						\$44	\$44	0%
	SHOPIFY	None	None	\$1,448	\$114					\$37	\$34	1%
Tradeline Subtype	Institution	Recurrence Type	Recurrence Frequency	Payment M1	Payment M2	Payment M3	Payment M4	Payment M5	Payment M6	Last Payment Amount	Avg. payment transaction size	Share of Payment
Non-Financial Tradelines	VERIZON	Both Amount & Time	Monthly	\$256	\$256	\$263	\$267	\$267	\$251	\$256	\$259	20%
	COMCAST CABLE COMM	Only Time	Monthly	\$271	\$271	\$271	\$271	\$271	\$271	\$271	\$271	21%
	SPRINT WIRELESS	Only Time	Monthly	\$1,479	\$274	\$274	\$274	\$274	\$274	\$1,479	\$274	37%
	VERIZON	Only Time	Monthly			\$302	\$433	\$433	\$450	\$302	\$322	21%
	BILL PAYMENT	Unknown	None		\$234					\$234	\$234	6%
AUTO INSURANCE	GEICO	None	None	\$264	\$264	\$260	\$260	\$260	\$260	\$148	\$132	100%

FIG. 2D

256		298		299		294	
VARIABLE DIMENSION & FEATURE ENGINEERING		SELECT ATTRIBUTES		Variable Name		Customer 1	Customer 2
Cash flow	Avg. daily balance, days cash in hand, min. balance			AutoLoanIndicator		1	1
				BusinessLoanIndicator		1	1
Customer Financials	Income/ Revenue, Debt to income, income concentration, forecasted next month income			CashInHandDays		31	0
				CoefVarCreditCount		16%	18%
Recurrence	Stability of income, recurring debt obligations, missed payment, recurring credit			CoefVarDebitCount		11%	7%
				CoefVarRevenue		17%	14%
Trends	Velocity, volatility, out of pattern, forecasts, seasonality			CommercialLoyaltyIndex		1	3
				CommercialRiskIndex		3	5
Financial ecosystem (Entity extraction and metadata enrichment)	Institution & vendor name, income source, Rent, Payroll providers, competitor intelligence, payment to collection agency, insurance claims deposit, luxury restaurant, New age lender			CreditCardAmountMonth1		\$161,074	\$1,672
				CreditTransactionsAmountMonth1		\$315,624	\$131,176
Negative Triggers	NSF, overdraft, missed payment, late payment fee			DebtRepaymentMonth1		\$47,255	\$10,784
				DollarATMCredit		\$10,220	\$0
Financial Tradelines	Tradeline type, invisible trades, new deposit, payment history, wallet size & share			DSCR1		10%	71%
				Electricity/Gas/PowerPaymentCount		4	0
Non-Financial Tradelines	Tradeline type- electricity, water, telecom, new deposit, payment history			GrossProfitL1byP1		65%	269%
				InsuranceIndicator		0	1
Index ¹	Risk index, loyalty index			MaxBalanceMonth1		\$363,893	\$14,460
				MaxOverdraftUtilizedMonth1		\$0	\$68,021
Benchmarking	Debt to income percentile, ATM withdrawal percentile			MerchantReversalsAmountMonth1		\$0	\$0
				MinBalanceMonth1		\$270,212	-\$17,032
Customer profile & life stages	Preferences (Digital savvy, traveler, etc.), life events			MOMTotalDebtRepayment1		72.7%	-12.7%
				MortgageIndicator		0	1
Debit card spend profile/ pattern	Discretionary spend, healthcare spend profile/ pattern			NSFCountTotal		2	46
				NSFLastDays		37	5
				NSFTotalAmount		\$70	\$1,288
				NumBranchCredit		21	0
				NumDigitalWalletDebitMonth1		0	0
				NumLoanCredits		9	3
				NumLoanCreditMonth1		1	0
				OverdraftLimit		\$0	\$20,000
				PayrollFrequency		others	others

FIG. 2E

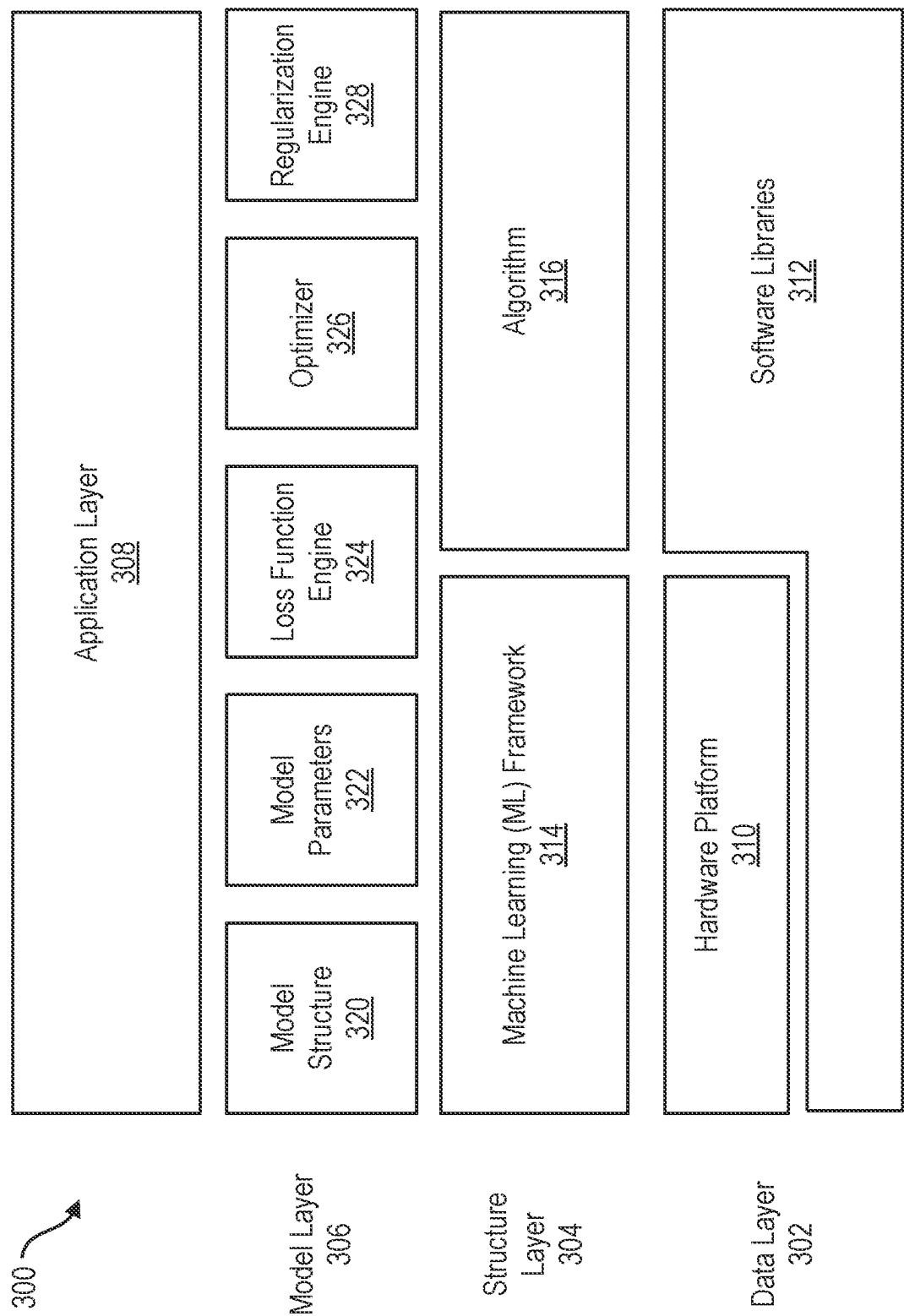


FIG. 3A

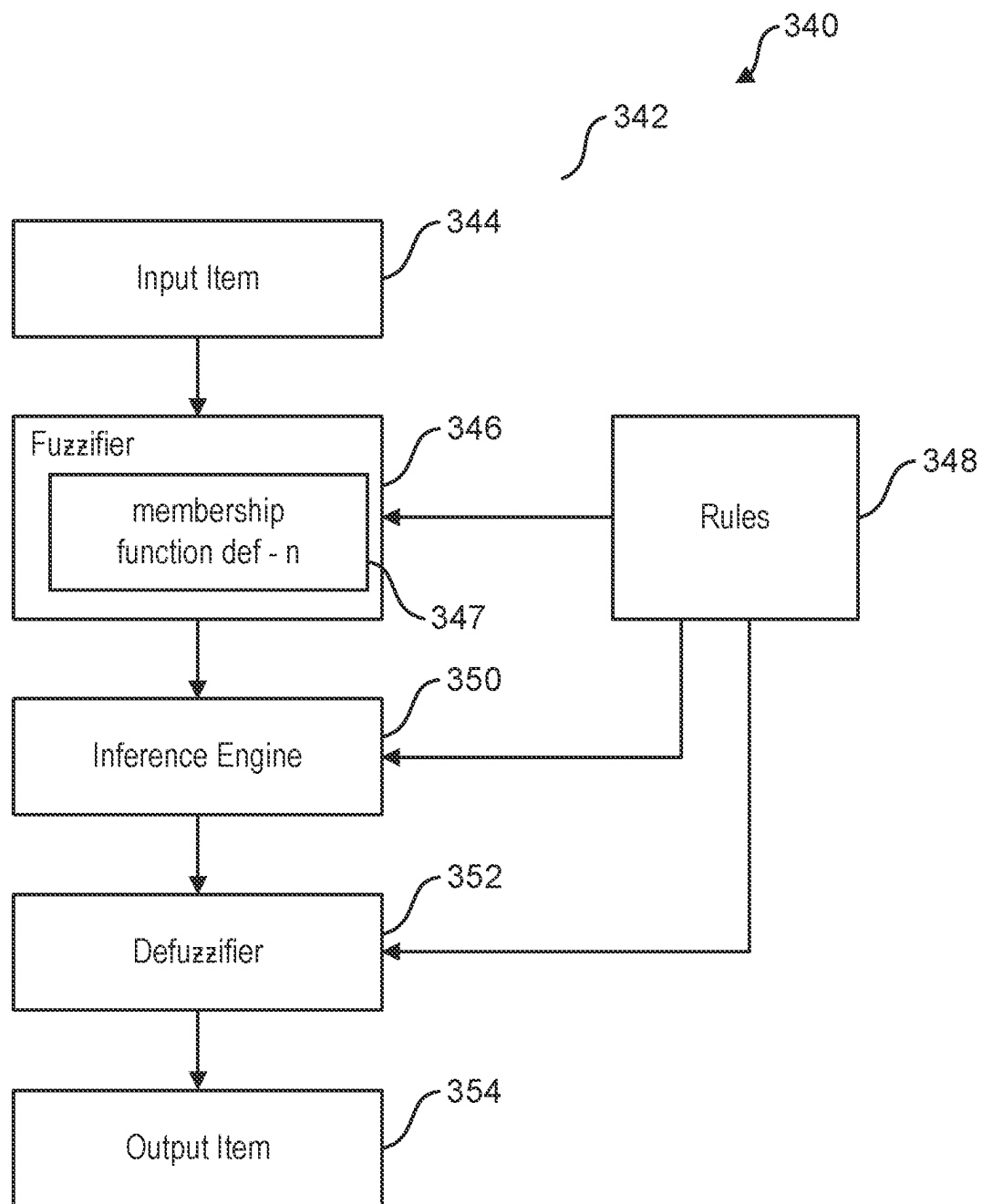


FIG. 3B

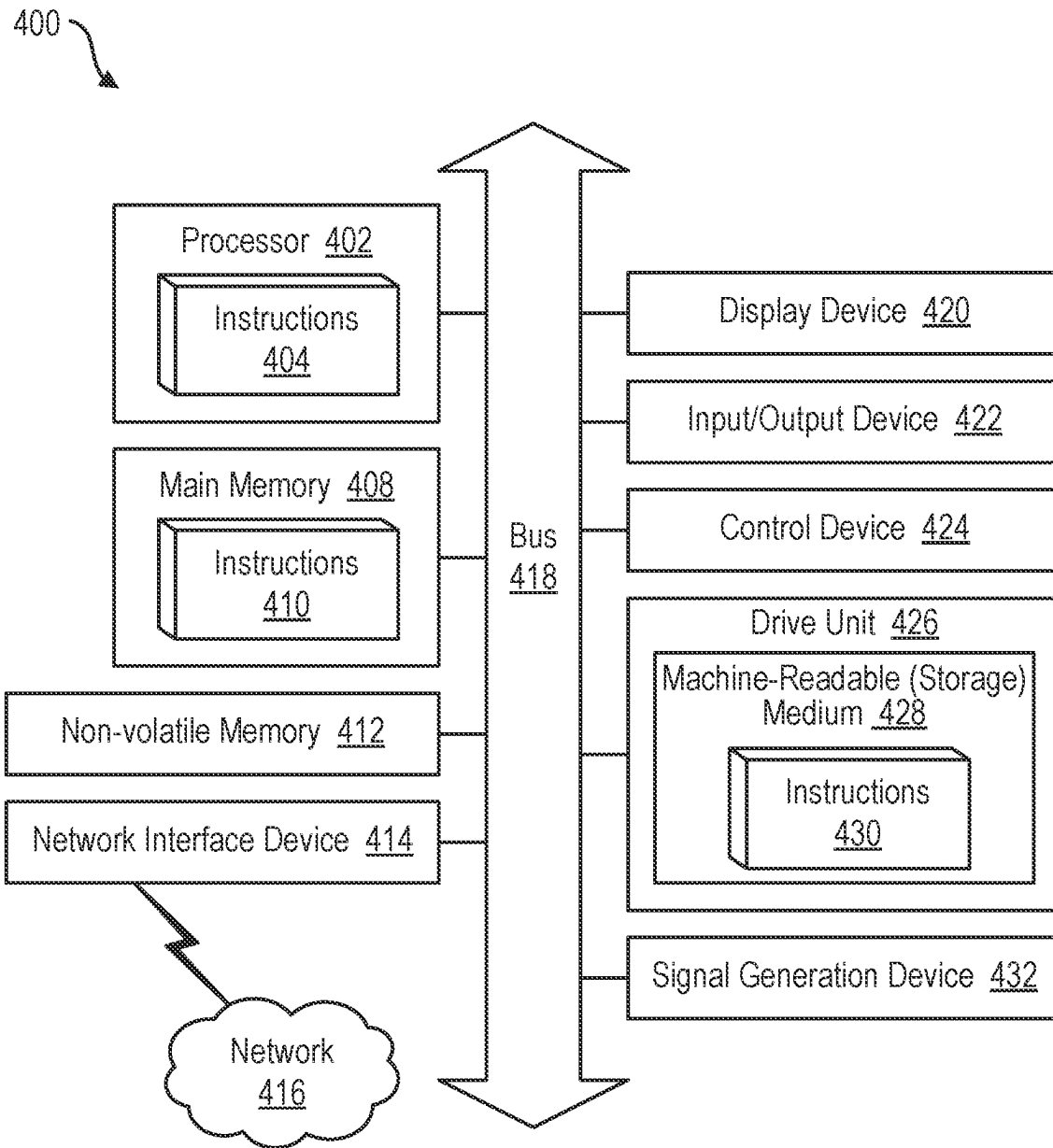


FIG. 4

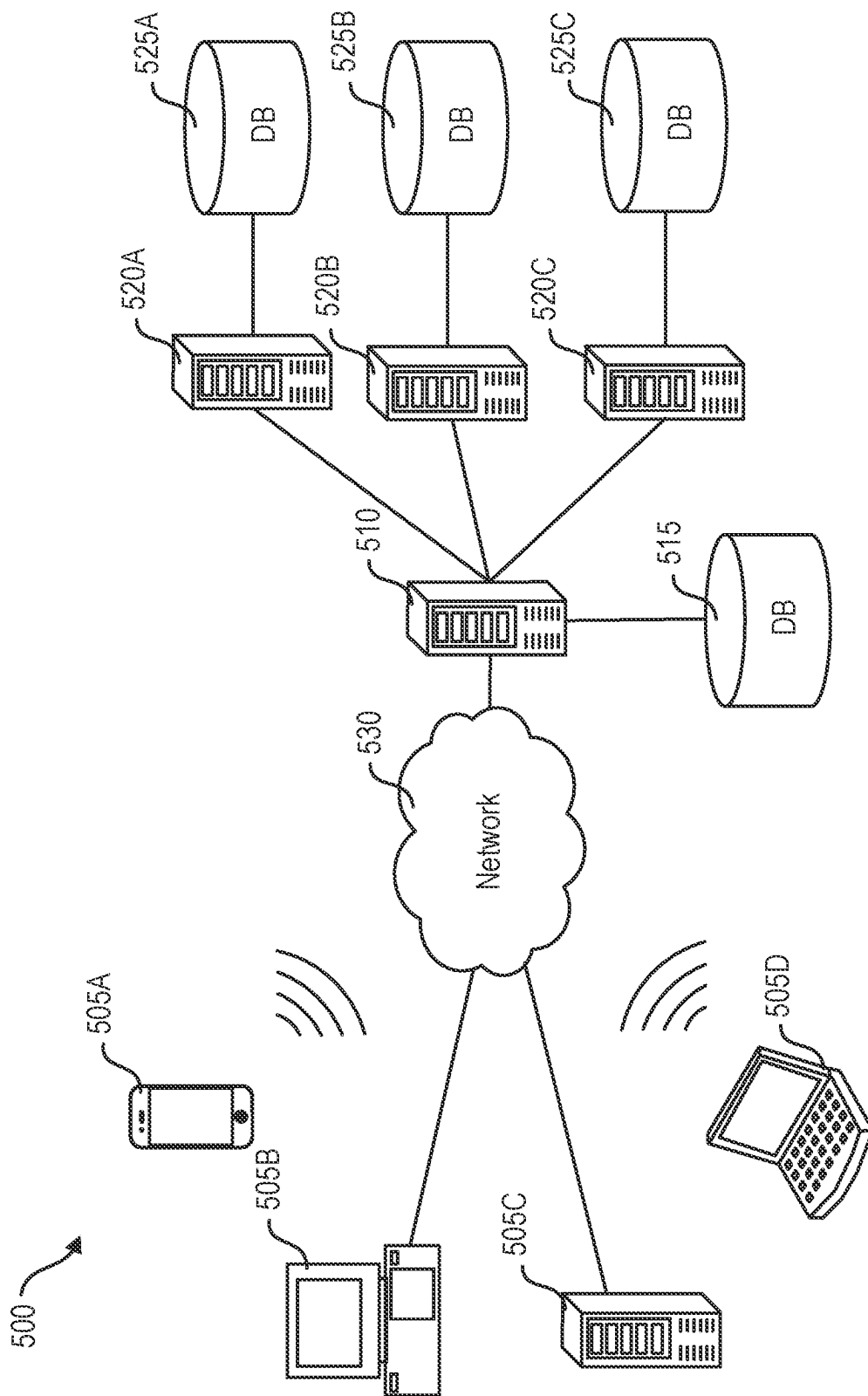


FIG. 5



ABC Inc.

- Retail industry
- Revenue from multiple sources
- Current checking account relationship with wellsFargo

Date	Amount	Transaction Description	Type
1/1/2020	\$ 2,200	ACH CREDIT SHOPIFY PAYMENTS TRANSFER	Credit
1/15/2020	\$ 3,100	ONLINE TRANSFER FROM CHECKINGXX01 ON 01/15 CUSTOMER REFERENCE/AMAZON	Credit
1/16/2020	\$ 5,400	BANK REFERENCE NQ:XXTJ GLOBAL DEP GLOBAL PAYMENTS CCD XX83	Credit
1/13/2020	\$ 4,000	EPAY MOBIL XXXXXX1234 CHASE CREDIT CRD COIODXXX1234 DIRECT DEBIT	Debit
1/16/2020	\$ 1,000	BANK OF AMERICA CREDIT CARD BILL PAYMENT	Debit
1/25/2020	\$ 980	ELECTRONIC/ACH DEBIT KABBAGE INC	Debit
1/31/2020	\$ 1,910	WELLS FARGO MORTGAGE XX5667 EB ID: XX7117	Debit
1/19/2020	\$ 202	DUKE ENERGY IN WEB_PAY JAN 20XX235	Debit
1/22/2020	\$ 252	VZ WIRELESS VE VZW WEB PAY XX77	Debit
1/23/2020	\$ 258	WITHDRAWAL GERBER LIFE INS - ID: XXXXXX1752 CO: GERBER LIFE INS	Debit
1/02/2020	\$ 258	EVERETT ST OFFICE RENT	Debit

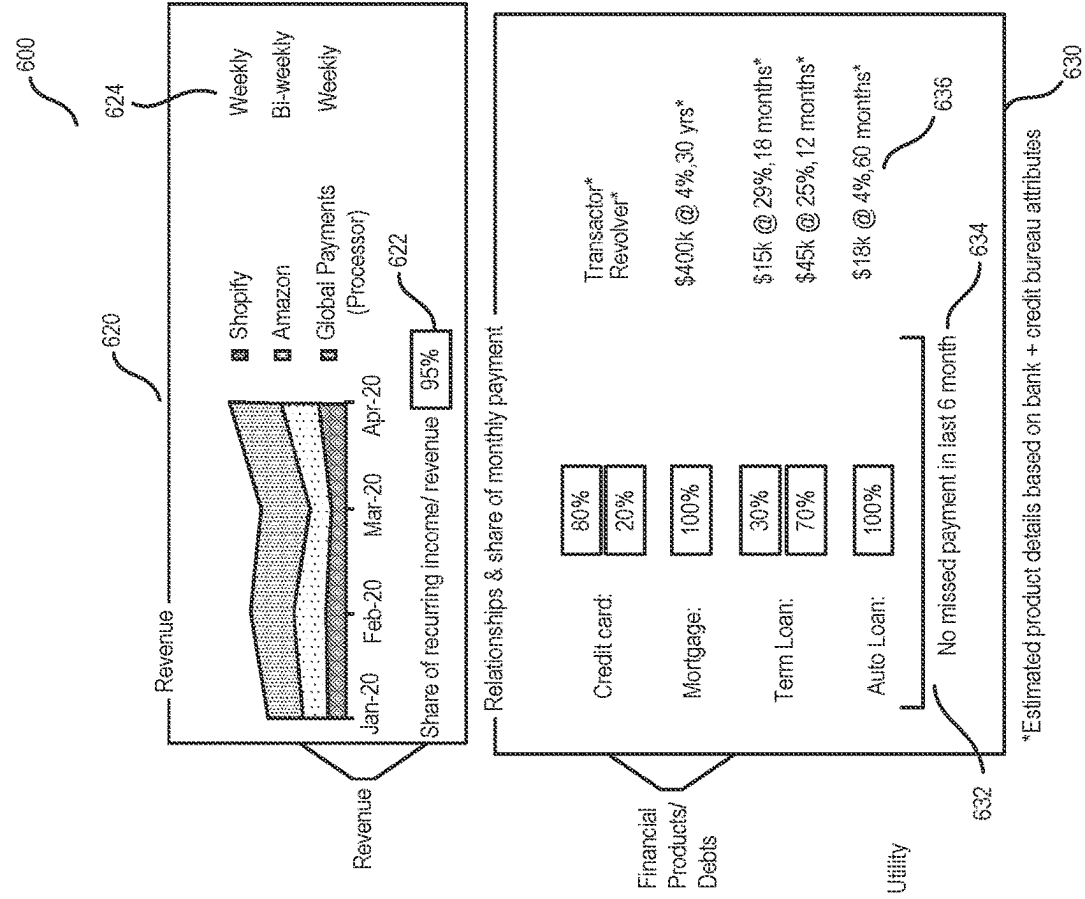
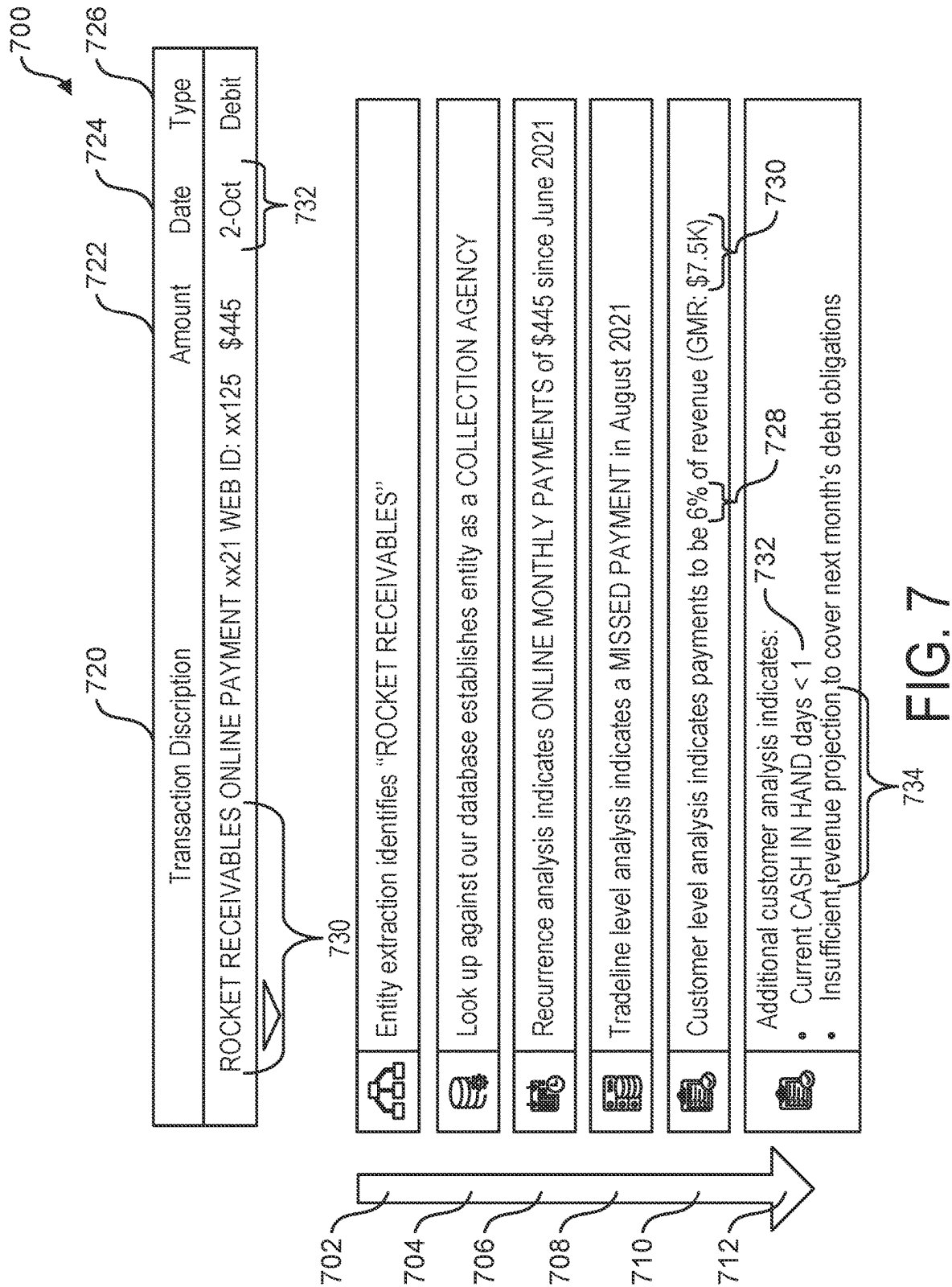


FIG. 6



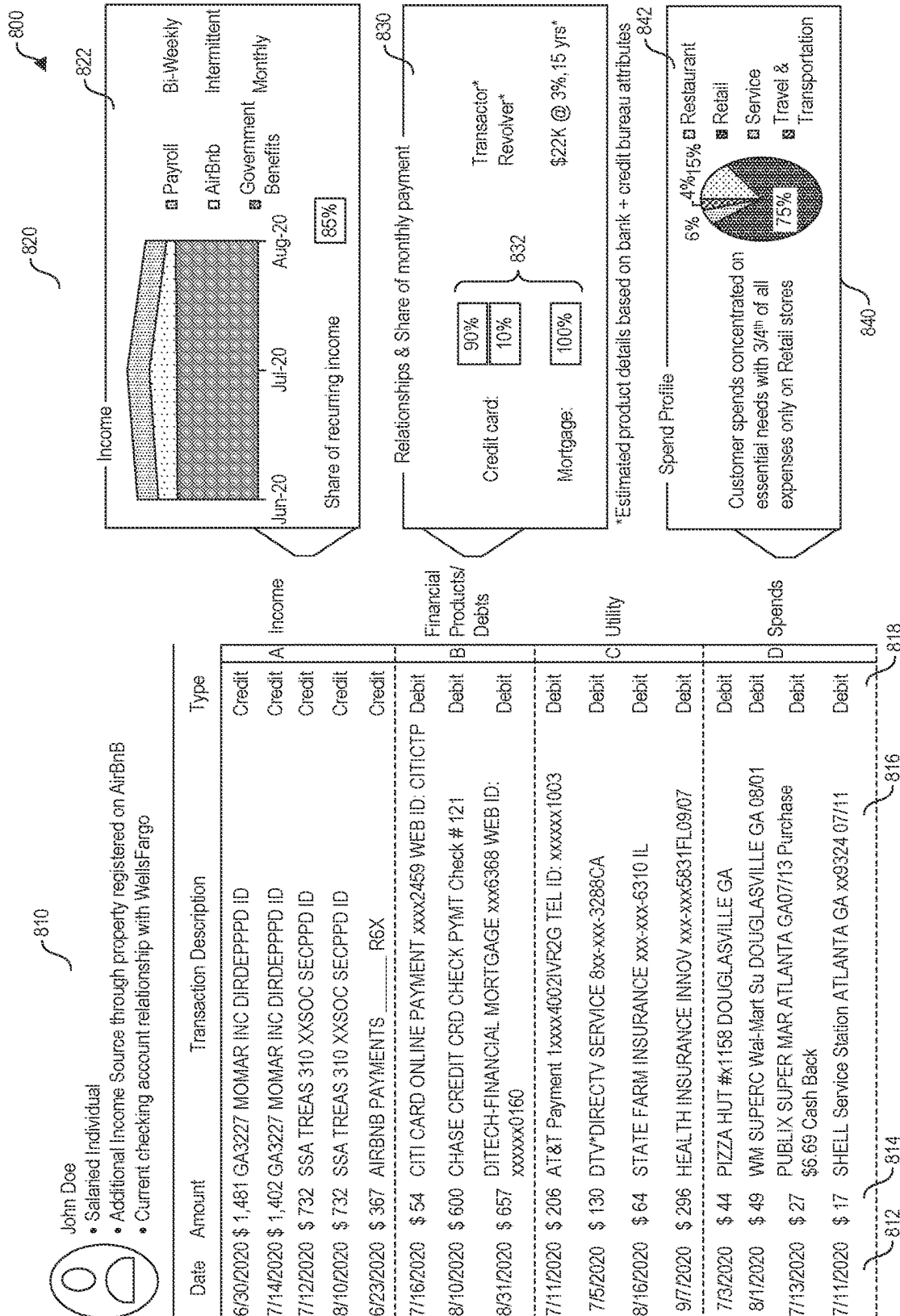
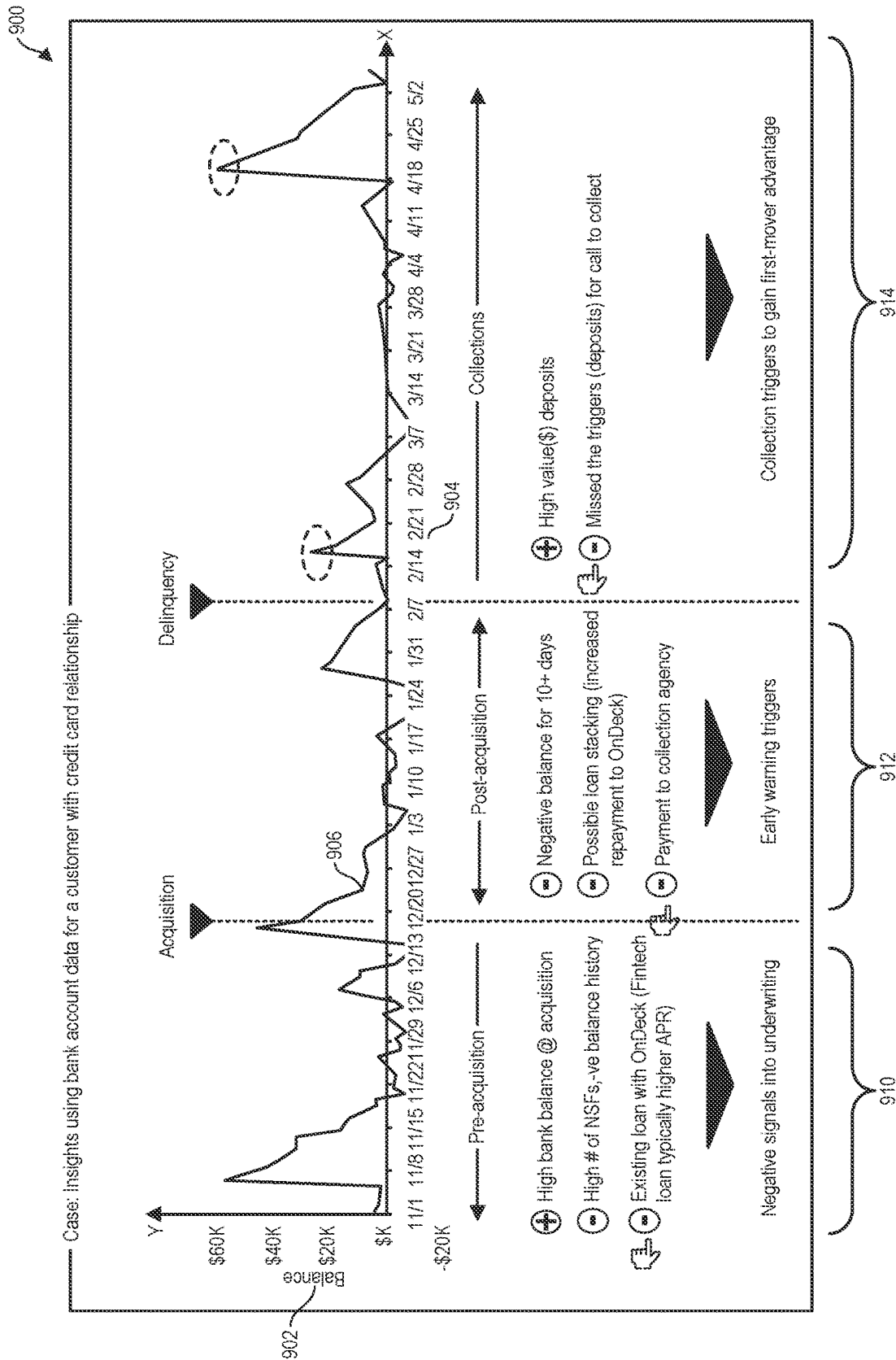


FIG. 8



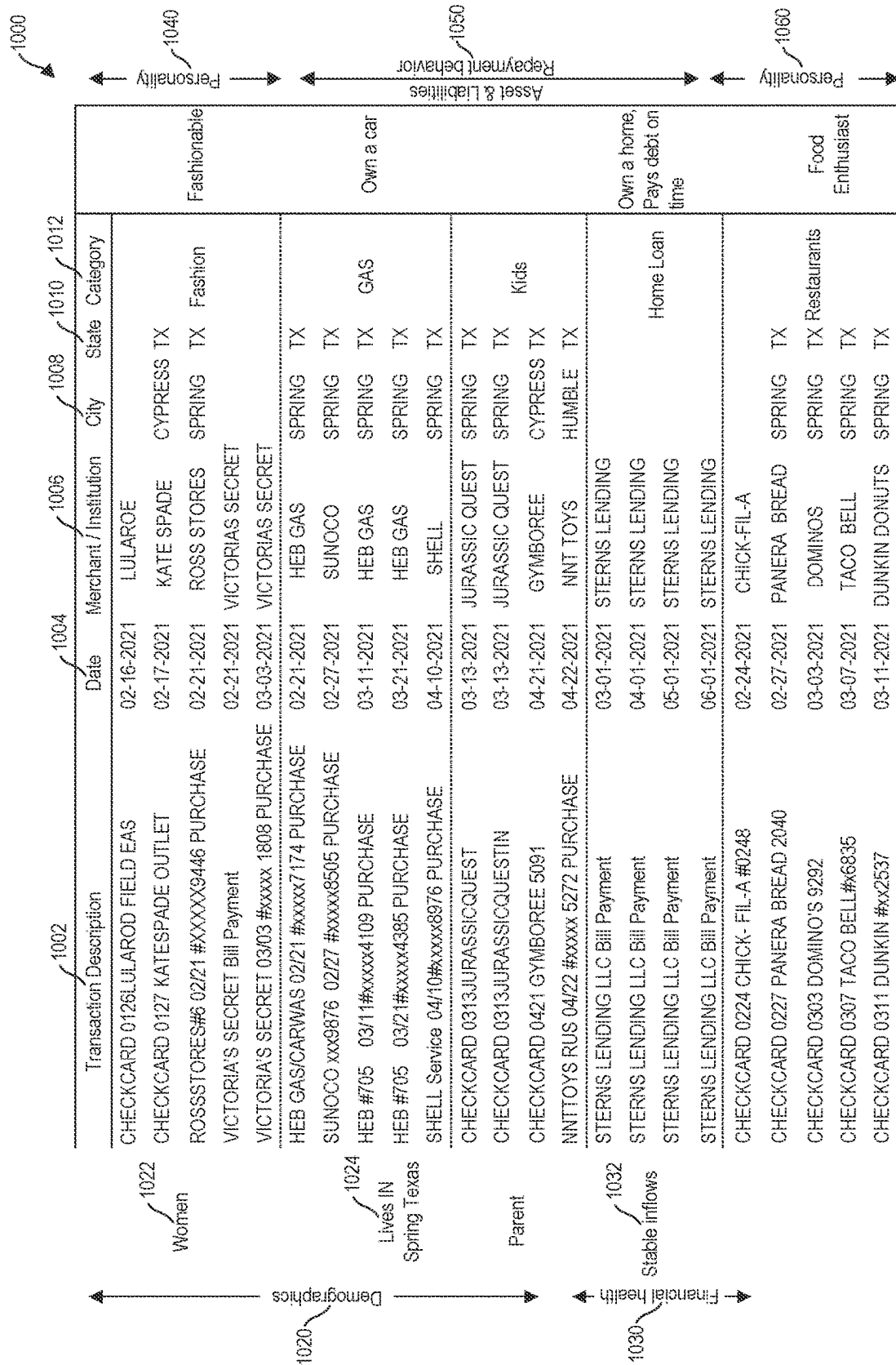


FIG. 10

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ARTIFICIAL INTELLIGENCE BASED TRANSACTIONS CONTEXTUALIZATION PLATFORM

TECHNICAL FIELD

The systems, methods, and computer-readable media disclosed herein relate generally to data analytics and contextualization platforms, such as artificial intelligence based transaction contextualization platforms. Some aspects of the data analytics and contextualization platforms disclosed herein relate to the use of fuzzy logic on unstructured input data to enable contextualization of various input data items that relate to the unstructured input data. Contextualization enables analysis of the input data items by downstream artificial intelligence/machine learning (AI/ML) models.

BACKGROUND

In transactional and data warehousing systems, datasets can include structured (e.g., delimited, addressable) and unstructured data. For structured data, data analytics systems can be used, alone or in conjunction with AI/ML models, to perform operations (e.g., calculations) across records in datasets. Levels of granularity for the calculations can be determined based on various structured data items in the datasets. For example, if several records contain numerical values that correspond to a particular value of a grouping variable, the numerical values can be summarized, across the several records, for the grouping variable. However, unstructured data does not lend itself to use as a grouping variable for data grouping operations. Additionally, feeding unstructured data, without pre-processing, to AI/ML models can adversely impact operating time and predictive value of these models and increase requirements for computing resources, such as memory and disk space. Additionally, some ML models may not natively be able to process unstructured data.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A shows an example computing environment that includes a data analytics and contextualization platform in accordance with some implementations of the present technology.

FIG. 1B shows example inputs of the data analytics and contextualization platform of FIG. 1A in accordance with some implementations of the present technology.

FIGS. 1C-1E show example published outputs (entity behavior maps) generated by the data analytics and contextualization platform of FIG. 1A in accordance with some implementations of the present technology.

FIG. 2A is a flowchart depicting an example method of operation of the data analytics and contextualization platform of FIG. 1A, in accordance with some implementations of the present technology.

FIGS. 2B-2E show example dashboards that visualize aspects of entity behavior maps generated by the data analytics and contextualization platform of FIG. 1A, in accordance with some implementations of the present technology.

FIG. 3A illustrates a layered architecture of an artificial intelligence (AI) system that can implement the machine learning models of the data analytics and contextualization platform of FIG. 1A, in accordance with some implementations of the present technology.

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FIG. 3B illustrates an example fuzzy logic engine of the data analytics and contextualization platform of FIG. 1A, in accordance with some implementations of the present technology.

FIG. 4 is a block diagram showing some of the components typically incorporated in at least some of the computer systems and other devices on which the data analytics and contextualization platform operates in accordance with some implementations of the present technology.

FIG. 5 is a system diagram illustrating an example of a computing environment in which the data analytics and contextualization platform operates in some implementations of the present technology.

FIGS. 6-10 illustrate example use cases of the data analytics and contextualization platform, in accordance with some implementations of the present technology.

The drawings have not necessarily been drawn to scale. For example, some components and/or operations may be separated into different blocks or combined into a single block for the purposes of discussion of some of the embodiments of the disclosed system. Moreover, while the technology is amenable to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and are described in detail below. The intention, however, is not to limit the technology to the particular embodiments described. On the contrary, the technology is intended to cover all modifications, equivalents and alternatives falling within the scope of the technology as defined by the appended claims.

DETAILED DESCRIPTION

Despite not being suitable for certain types of machine analytics in its raw form, unstructured data can contain items or pointers to valuable contextual information that help clarify items in the input data. Accordingly, described herein is a data analytics and contextualization platform that enables contextualization of input data items by extracting meaning from unstructured data items that are related to the input data items (e.g., included in the same record, dataset, set of key-value pairs, document, and so forth).

Contextualization of input data enables a host of technical advantages. For example, machine-generated (synthetic) contextual data can be provided, along with context-relevant items extracted from input data, to downstream AI/ML systems in the form of an enriched set that includes features not directly available in raw input data. The enriched features can enable AI/ML models to provide targeted and relevant insights and recommendations. Compared to the size of a raw input dataset, generating enriched features that contain a predetermined set (e.g., 1-100) of contextual items and their corresponding value(s) can reduce the size of the input dataset, which can increase processing speed and model efficiency. Furthermore, performing certain feature generation operations in parallel, as described herein, reduces overall time of data pre-processing and feature generation. Contextual items can help optimize the overall pre-processing time, serving as beginning markers in dynamically constructed dependency trees for parallel processing in summarization operations.

In operation, the platform acquires an input dataset from one or more source computing systems. According to one use case, the input dataset can include transaction data and free-form text items, such as transaction description. The platform can perform various pre-processing operations to cleanse and normalize items in the input dataset. The plat-

form then extracts textual data from one or more unstructured, free-form text items in the input dataset.

Using fuzzy logic, the platform generates a set of contextual items based on the unstructured, free-form text. Contextual items can be thought of as synthetic items not readily available in the input dataset and inferred or generated based on data in the input dataset. For example, the platform can extract from the free-form text or generate items that provide further insight into records in the input dataset and allow the items in the input dataset to be labeled (tagged, associated with, appended to, and so forth) or included in particular features such that the items can be analyzed using AI/ML models. For example, in a transaction-related context, contextual items can include various synthetic properties generated or determined for a particular transaction record, where the properties are not otherwise available in any of the structured data fields associated with a particular transaction record. These synthetic properties can include, for example, various generated classifiers for a transaction, which describe the context for a transaction. For example, the transaction can be an automatic direct deposit payment, which can be determined by using a fuzzy logic algorithm on an item in the transaction description field to reference and extract an item/token that likely corresponds to a record in an employer ontology. In another example, a transaction can be a debt payment transaction as determined, for example, based on transaction periodicity in combination with using a fuzzy logic algorithm to reference and extract an item/token that likely corresponds to a record in a creditor ontology. In another example, the transaction can be determined to be a bill payment transaction, discretionary spending transaction, and so forth. Other synthetic properties of contextual items can include transaction recurrence indicators (amount, time, type, frequency), entity identifiers, merchant identifiers, transaction channel identifiers, entity or transaction location identifiers, and so forth.

Using the outputs of contextualization operations (for example, by referencing labels on the input records), the platform can generate entity behavior maps. The entity behavior maps can show various roll-ups of items in the input dataset based on the determined contextual items. For example, entity behavior maps can provide insight into a particular entity's behavior at the transaction-, tradeline-, or customer/entity level. The platform can perform summarization operations for items on the generated entity behavior maps to generate various aggregations relating to structured items included in the input data and/or contextual items generated based on unstructured input data. The type of aggregations or operations to perform can be determined by the type of the corresponding entity behavior map. For example, certain features, such as a risk index, can be included at the tradeline- or customer-level but not at the transaction level. The platform can further perform higher-level insight generator operations to generate scores, indexes and so forth, particularly at the higher rollup levels, such as tradeline and/or customer. Example types of entity behavior maps (transaction-, tradeline-, or customer/entity level) and their corresponding calculated items are shown with respect to FIGS. 1C, 1D, and 1E.

The platform can publish the output(s) of the above operations for consumption by one or more target computing systems (e.g., in the form of application programming interface (API) messages to one or more target applications). These output items can be included in tabular or key-value pairs, and are also shown with respect to items 220 of FIG.

2B as, for example, classification items 224a-c, amount 226a, time 226b, frequency 226c, entity/merchant 228, channel 230, location 240.

Example Embodiments of a Data Analytics and Contextualization Platform

FIG. 1A shows an example computing environment 100 that includes a data analytics and contextualization platform 110 in accordance with some implementations of the present technology.

As shown, the environment 100 includes one or more of a source computing system 102 and one or more of a target computing system 104. These systems can be communicatively coupled to the data analytics and contextualization platform 110 via a network. Each of the source computing system 102, target computing system 104, and data analytics and contextualization platform 110 can each include various components, including one or more processors, memory modules, transceivers, network interfaces, databases, executable files (in binary form and/or in compiled form), libraries of executables, file structures, and so forth.

In some implementations, any of the source computing system 102, target computing system 104, and data analytics and contextualization platform 110 can be distributed across more than one computing devices. For example, a particular instance of the data analytics and contextualization platform 110 can be deployed as an executable environment available to a subscriber entity (e.g., an entity associated with a particular target computing system 104) in a cloud-based environment, such as, for example, in a virtual private cloud, via a virtual network, in DaaS (data-as-a-service) computing environment, SaaS (software-as-a-service) computing environment, PaaS (platform-as-a-service) computing environment, IaaS (infrastructure-as-a-service computing environment) and/or the like. Accordingly, the executable environment can be deployed as a container, pod of containers, cluster of containers, or a dedicated computing grid in a cloud-based environment, which provide varying levels of process and data isolation to meet various levels of data privacy and regulatory standards. At a minimum, the cloud-based implementation infrastructure described herein allows (at the container level) for isolating API calls and data workflows, which secures and isolates data streams and data stores of a particular entity (e.g., an entity associated with a particular source computing system 102 or target computing system 104).

An example source computing system 102 can be or include any suitable computing system that provides transactional or warehoused (aggregated, dimensioned) data, such as a financial institution computing system, an insurance claims management system, an underwriting system and so forth. A target computing system 104 can be or include a computing system associated with a consumer entity for the output data of the data analytics and contextualization platform 110, which may be the same or different entity relative to the source computing system 102. For example, a particular source computing system 102, connected to a particular instance of the data analytics and contextualization platform 110, can be operated by a particular financial institution and include transaction data. The financial institution can also be a consumer of the outputs of the particular instance of the data analytics and contextualization platform 110 via a particular target computing system 104. In another example, another particular source computing system 102, connected to the particular instance of the data analytics and contextualization platform 110, can include a computing system of a financial transaction processor or aggregator (e.g., and entity that provides data via

Open Banking or a similar communications infrastructure, where the data is contributed by ecosystem participants, such as in a consortium fashion), and the financial institution can consume the outputs of the data analytics and contextualization platform **110** via the particular target computing system **104**, wherein the outputs are generated using aggregated financial transaction data from multiple financial institutions, using transaction data for a customer of the financial institution, and so forth.

As shown, the data analytics and contextualization platform **110** includes various engines, some of which can be omitted or combined according to various implementations. As used herein, the term “engine” refers to one or more sets of computer-executable instructions, in compiled or executable form, that are stored on non-transitory computer-readable media and can be executed by one or more processors to perform software- and/or hardware-based computer operations. The computer-executable instructions can be special-purpose computer-executable instructions to perform a specific set of operations as defined by parametrized functions, specific configuration settings, special-purpose code, and/or the like. The engines can generate and/or receive various electronic messages (e.g., via channels **111**, **121**). Whenever a particular message is referred to in singular form, one of skill will appreciate that more than one electronic messages can be used to carry out the described operations. For example, a particular dataset, record, or item therein can be broken down into multiple messages. Furthermore, a particular system or module can generate or receive multiple items (e.g., datasets, records, and/or items) in a particular message.

As shown according to an example implementation, the various engines of data analytics and contextualization platform **110** can include a data interceptor **120**, a preprocessor **130**, a contextualizer **140**, a feature definition engine **145**, a summarizer **150**, a processing pipeline orchestrator **155**, an insight generator **160**, and a publisher **170**.

The data interceptor **120** executes operations to acquire (receive, query, and so forth) input datasets from one or more source computing systems **102**. In some implementations, the input data can be acquired via queries from various data sources associated with source computing systems **102**, such as one or more of a data lake **102a** and/or a database **102b**. For example, a particular input dataset can originate from a data warehousing system, such as a data lake **102a** and/or from a data warehousing or transactional database **102b**. In some implementations, the input data can be received from a particular application **102d**, in the form of an API message, at the API endpoint **122** of the data interceptor **120**. The API endpoint **122** can be a specific location within an API that accepts requests and sends back responses. In some implementations, the input data can be received, at the storage resource **124** of the data interceptor **120**, from a particular file system **102c** and/or from a particular application **102d** (e.g., via an FTP process or another similar process). The storage resource **124** (e.g., an Amazon Simple Storage Service (S3) bucket or similar) can be a cloud storage resource that receives and stores various files in an electronic data interchange ecosystem. The API endpoint **122** and/or the storage resource **124** can be an addressable location on a physical or virtual server of the platform **110**. The addressable location can be identified by a URL, and IP address, a port number, or a combination thereof.

FIG. 1B shows example inputs **192** of the data analytics and contextualization platform of FIG. 1A. A particular input dataset can be included in an API message and can

include one or more transaction records (**192a**, **192b**, **192c**). The transaction records can include various items, such as transaction identifiers, customer identifiers, account identifiers, transaction types (e.g., credit, debit), descriptions, amounts, date/time stamps, account available balance, and so forth. These items are also shown with respect to items **220** of FIG. 2B as, for example, items in example transactions **221** (e.g., description **222a**, amount **222b**, date **222c**). One of skill will appreciate that the illustrated inputs are merely examples, and the data analytics and contextualization platform **110** can be utilized in other contexts, such as insurance claim analytics, underwriting, and so forth.

The preprocessor **130** executes pre-processing operations to optimize and/or normalize items in the input dataset. The pre-processing operations can include renaming items in the received dataset in accordance with a particular naming convention or ontology (e.g., from a base feature store **145a**, custom feature store **145b**, and so forth), checking and resolving unexpected values, and so forth. The pre-processing operations can further include performing quantization operations (e.g., data type conversion, such as from a floating point value to an integer value, aimed to reduce the size of input data and improve model efficiency), replacing null values with default values, filling in missing data using regression-based machine learning models, concatenating and/or truncating string values, segmenting long text items into a plurality of strings of predetermined length, and so forth. To that end, the preprocessor **130** can include an AI/ML engine **132** structured to perform AI/ML based preprocessing operations, such as filling in missing data using regression algorithms, segmenting (tokenizing) text items, and so forth.

The contextualizer **140** executes operations to enable contextualization of input data items and make input data items suitable for analysis via artificial intelligence/machine learning (AI/ML) downstream of the contextualizer **140** (e.g., the AI/ML engines **144**, **146**, **148**, **152**, and/or **162**). To that end, the AI/ML engine **142** can execute fuzzy logic and/or elastic search operations on an item in the input dataset (e.g., a free-form text field, such as transaction description) and/or on one or more tokens extracted therefrom, as described, for example, relative to FIGS. 2A and 3B. The contextualization operations of the AI/ML engine **142** generate one or more contextual items. Contextual items enable various transaction insights and can serve as a starting point for generating, by the summarizer **150**, of various entity behavior maps. In a transaction use case, contextual items can include, for example, classification items (income/revenue transaction class descriptors, debt payment transaction class descriptors, bill payment transaction class descriptors, spend transaction class descriptors, and/or fee transaction class descriptors), recurrence items (amount, time, type, frequency), entity/merchant, channel, location, and so forth.

In some implementations, the contextualizer **140** can perform various additional operations using the input dataset and/or generated contextual items. For example, the entity extractor **144** of the contextualizer **140** can determine various entities based on the input dataset. The entities can correspond to various levels at which entity behavior maps can be generated, such as transaction, tradeline, and/or customer. Some entities can be determined by parsing items from the input dataset. With reference to an example transaction record **192a** of FIG. 1B, transaction-level data can be identified in the input dataset by “Transaction_ID” or a similar item that includes a transaction identifier. Customer-level data can be identified in the input dataset by “Custom-

er_ID", "Account_ID" or a similar item that includes a customer unit or entity identifier. Other entities, such as tradelines, can be determined using the generated contextual items, as described according to an example of FIG. 2A. As another example, the classifier **146** of the contextualizer **140** can execute a classification machine learning model to analyze the generated contextual items, such as merchant identifiers, dates, debit or credit transaction type, channels, locations, to determine how the contextual items relate to various transaction class descriptors. As yet another example, the recurrence identifier **148** of the contextualizer **140** can generate a dataset that includes a subset of transactions that correspond to a particular contextual item and execute operations to infer recurrence patterns by determining date differences (e.g., in days, months, and so forth) between any two transactions in the set, where the transactions are ordered to occur one after another without intervening transactions of similar type.

The summarizer **150** executes operations to generate various statistical summaries (e.g., sums, averages, lowest values, highest values, and so forth) for corresponding to particular contextual items and at particular roll-up levels. The contextual items can be thought of as columns or items in a resulting output dataset. The particular roll-up levels can be thought of as a particular type of output dataset/entity behavior map (e.g., transaction-level, tradeline-level, customer-level), which can be included in metadata associated with an outbound API message for the output dataset. A particular roll-up level, therefore, can correspond a particular type of entity behavior map **154** on which the API message is based. To that end, the summarizer **150** can generate one or more entity behavior maps **154** and then execute (e.g., at the tradeline and/or customer level) various summarization operations to populate items in the entity behavior maps **154**. The particular summarization operations can be selected based on the type of an entity behavior map such that the corresponding executables or AI/ML models are conditionally activated based on the determined type of entity behavior map. For example, certain variables, such as customer credit risk, can be estimated at the customer level rather than at the transaction level and therefore AI/ML models that make these determinations would not be activated for transaction data. This approach, advantageously, can improve performance of the activated AI/ML models by providing to the models, as appropriate, feature maps that include data aggregations rather than, or in addition to, raw data. For example for certain items, raw data (e.g., amounts) may be less meaningful than scaled data (e.g., percentage of on-time payments across accounts).

FIGS. 1C-1E show example published outputs (corresponding to entity behavior maps **154**) generated by the summarizer **150**.

A transaction-level entity behavior map **154** of FIG. 1C includes example items **194**, which describe how a particular entity (commercial or consumer) functions at a transactional level of granularity. The items can include values parsed from input data (e.g., transaction id, description, amount, date) and/or synthetic contextual items generated or determined based on input data (e.g., customer id, account id, transaction type, transaction category, vendor name, institution name, recurrence type, recurrence frequency, auto-pay yes/no indicator, channel, city).

A tradeline-level entity behavior map **154** of FIG. 1D includes example items **196a** and/or **196b**, which describe how a particular entity (commercial or consumer) functions at a tradeline level of granularity. The items can include values parsed from input data and/or synthetic contextual

items generated or determined based on input data (e.g., customer id, tradeline id, tradeline type, tradeline subtype, institution (e.g., lender, counterparty, merchant) name, institution type, recurrence frequency, periodicity information, and so forth). The items can include various aggregations (e.g., examples of tradeline analytics) generated by the summarizer **150**, such as periodic payment amounts (payment ticket size), actual payment amounts, last payment date/time, last payment amount, and/or share of payment. Additional examples of tradeline-level data (**272**, **274**) are shown with respect to FIG. 2D. For example, a transaction or group of transactions in a particular tradeline (**272**, **274**) can be further associated with a tradeline subtype **276**, contextual item **278** (e.g., institution, merchant), recurrence type **280**, recurrence frequency **282**, payment patterns **284**, determined last payment amount **286**, determined average payment transaction size **288**, and determined share of total payment **270**.

A customer-level entity behavior map **154** of FIG. 1E includes example items **198**, which describe how a particular entity (commercial or consumer) functions overall. The items can include values parsed from input data and/or synthetic contextual items generated or determined based on input data (e.g., customer id, auto loan y/n, business loan y/n, and/or payment frequency). The items can include various aggregations generated by the summarizer **150**, such as cash-in-hand days, credit count, debit count, debt repayment amounts, non-sufficient funds statistics, overdraft limit, and so forth. The items can include various scores/indexes **164**, smart variables **166**, and/or indicators **168** generated by the insight generator **160**, such as commercial loyalty index, commercial risk index, and so forth.

Referring to elements of FIG. 1A, the processing pipeline orchestrator **155** executes operations to synchronize operations of the contextualizer **140**, summarizer **150**, and/or insight generator **162**. For example, the processing pipeline orchestrator **155** can generate a dependency tree for the various operations performed by these engines and cause the engines to execute their respective operations in accordance with the dependency tree in order to minimize the overall execution time of the pipeline (e.g., from data ingestion to execution of publisher **170** operations).

In an example implementation, the processing pipeline orchestrator **155** can order certain operations and/or execute certain operations in parallel. Various contextual items can serve as starting point markers (triggers) in dynamically generated dependency trees. For example, once a particular contextual item is generated, the processing pipeline orchestrator **155** can cause the summarizer **150** to generate and/or update a particular entity behavior map **154** to include the contextual item and generate the corresponding summary statistics. In another example, various operations of the entity extractor **144**, classifier **146**, and recurrence identifier **148** can be executed in parallel such that a particular entity behavior map **154** is generated and/or updated to include additional columns as the items become available in the form of outputs of the corresponding entity extractor **144**, classifier **146**, and recurrence identifier **148**. For example, if the operations of classifier **146** complete first, the entity behavior map **154** can be first updated to include a first column for a particular generated transaction class descriptor. If the operations of recurrence identifier complete next, the entity behavior map **154** can be next updated to include second column(s) for the generated recurrence type, recurrence frequency, autopay y/n, periodicity, and so forth.

The contextualizer **140**, summarizer **150**, and/or insight generator **160** can work in conjunction with the feature

definition engine **145**, which can include a base feature store **145a** and/or a custom feature store **145b**. The feature definition engine **145** can execute operations to identify, using the input dataset and/or contextual items, a primary base variable, which can be a particular contextual item (e.g., item **298** of FIG. 2E) or a larger category associated therewith (e.g., dimension **256** of FIG. 2E). For example, a particular primary base variable can correspond to a determined debt payment transaction class descriptor, such as “Debt-Auto Loan”, which can be a contextual item generated based on a transaction description using fuzzy logic. The feature definition engine **145** can execute operations to relate the primary base variable to various categories (e.g., consumer, commercial, both), various risk levels (e.g., low, medium, high), data transformation operations (e.g., function calls that perform a particular aggregation operation (min, max, average) or calculation (sum, ratio) for a corresponding amount), and so forth. Accordingly the feature definition engine **145** can accept, as an input, a particular contextual item and a corresponding amount and return a calculation or aggregation (or perform another suitable AI/ML operation as described herein) by performing a function call selected based on the particular contextual item. For example, once a particular transaction is labeled with a particular contextual item, the contextual item for the transaction can be used to determine how to transform input data from the transaction record (e.g., amount) for inclusion in a particular feature.

In some implementations, the feature definition engine **145** provides output definitions (feature definition criteria) for higher-level operations of the insight generator **160**. Example output definitions **296** (shown in FIG. 2E) can be based on various items from entity behavior maps **154** and can be used to further augment the entity behavior maps **154**. In an example implementation, an AI/ML engine **162** of the insight generator **160** can accept as inputs various attribute items **298**, which can be structured as items (columns) in a particular entity behavior map **154**. The AI/ML engine **162** can use, for example, a neural network to assign weights to various input items and utilize the values the input items in accordance with the weights to generate the resulting output definitions. As an example, a particular dimension in the output can be conceptualized as cash flow, and its contributing attributes can include an average daily balance, minimum cash on hand, minimum account balance, and the like. Various customers/entities can be compared at **299** according to the resulting taxonomy.

The publisher **170** executes operations to generate one or more output datasets (e.g., the outputs of FIGS. 1C-1E), which can be received or otherwise accessed by an application **104a** of a target computing system **104**. In some implementations, the output datasets are included in API messages transmitted to the target computing system **104**. Example Methods of Operation of the Data Analytics and Contextualization Platform

FIG. 2A is a flowchart depicting an example method **200** of operation of the data analytics and contextualization platform **110** of FIG. 1A, in accordance with some implementations of the present technology. FIGS. 2B-2E show example dashboards that visualize aspects of entity behavior maps **154** generated by the data analytics and contextualization platform **110** of FIG. 1A, in accordance with some implementations of the present technology.

As a general overview, the method **200** includes acquiring an input dataset and performing pre-processing operations on the input dataset. The method **200** further includes extracting textual data from the input dataset and using

fuzzy logic to generate a set of contextual items, such as classification items, recurrence items (amount, time, type, frequency), entity/merchant, channel, location, and so forth. The method **200** further includes performing various entity contextualization operations and generating one or more entity behavior maps using the outputs of contextualization operations. The method **200** further includes performing summarization operations for items on the generated entity behavior map(s), performing insight generator operations, and publishing the output(s) of these operations. These items are discussed in accordance to example implementations, and one of skill will appreciate that various operations can be combined and/or omitted.

At operations **202**, the data interceptor **120** of the data analytics and contextualization platform **110** of FIG. 1A acquires an input dataset. In an example implementation relating to transaction analytics, the input dataset can include various raw transaction data elements, such as the elements included in electronic messages generated according to various financial data interchange standards (Open Banking, ACH, and so forth). The raw transaction data elements can include any or all of the input items described in reference to FIG. 1B, such as transaction identifiers, customer identifiers, account identifiers, transaction types, transaction description, transaction amounts, transaction dates, account available balance, and so forth. The dataset can be in a suitable structured format, such as Excel, json, or another tabular format. In some implementations, the input dataset is acquired via an API endpoint **122**. In some implementations, the input dataset is acquired via a storage resource **124**.

At operations **204**, the preprocessor **130** of the data analytics and contextualization platform **110** of FIG. 1A performs pre-processing operations on the input dataset. The pre-processing operations can include renaming items in the received dataset in accordance to a particular naming convention or ontology, checking and resolving unexpected values, and so forth. The pre-processing operations can further include performing quantization operations (e.g., data type conversion, such as from a floating point value to an integer value, aimed to reduce the size of input data and improve model efficiency), replacing null values with default values, filling in missing data using regression-based machine learning models, concatenating and/or truncating string values, segmenting long text items into a plurality of strings of predetermined length, and so forth.

At operations **206**, the contextualizer **140** of the data analytics and contextualization platform **110** of FIG. 1A generates a set of one or more contextual items using data from the input dataset. Generating the contextual items can include executing fuzzy logic operations with respect to one or more tokens from the input dataset, as described in reference to FIG. 3B. Using fuzzy logic, the data analytics and contextualization platform **110** is enabled to generate contextual items (e.g., merchant, periodicity) and use these items to contextualize various other items (e.g., amount) in the input dataset based on unstructured data from the input dataset. For example, contextual items can include classification items, recurrence/periodicity items (amount, time, type, frequency), entity/merchant, channel, location, and so forth. The generated contextual items are synthetic values that can make items in the input dataset suitable for analysis by downstream AI/ML engines (e.g., **144**, **146**, **148**, **152**, **162**) to generate various calculations, aggregations, summaries, predictions, and so forth.

In some implementations, the input dataset can contain a free-form text field, such as description **252** shown in

reference to FIG. 2C. The preprocessor **130** and/or contextualizer **140** can parse the free-form text field into one or more tokens. The tokens can correspond to units of information, such as sentences, words, strings of predetermined length such as a standard record length, and so forth. More generally, tokens can be extracted from or generated, for example, as described above and/or in U.S. patent application Ser. No. 17/988,684, which is incorporated herein by reference in its entirety and for all purposes. In some implementations, tokens correspond to segments of the description field, where the segments have predetermined start and end positions. In some implementations, tokens are identified as textual content that follows a particular label (e.g., “DES”, “PMT ID”, “ID”) and delimiter (e.g., “:”, “ ”, “,”). In some implementations, the entirety of the free-form text field constitutes a token.

In some implementations, the tokens are further labeled with metadata that describes various features determined by automatically processing the tokens. The features can describe various determined properties of the tokens, such as item length, item location (within a larger free-form text), presence of a particular character sequence, and so forth. The metadata makes the tokens suitable for processing and classification using fuzzy logic, such as by the fuzzifier component of an example fuzzy logic engine described in relation to FIG. 3B.

At operations **208a**, **208b**, and/or **208c**, various entity contextualization operations can be performed by engines of the contextualizer **140** using the generated contextual items and/or items extracted from the input dataset.

For example, the contextualizer **140** can, at **208a**, extract items from the input dataset using the entity extractor **144**. The entities can correspond to various levels at which entity behavior maps can be generated, such as transaction, tradeline, and/or customer. With reference to an example transaction record **192a** of FIG. 1B, transaction-level data can be identified in the input dataset by “Transaction_ID” or a similar item that includes a transaction identifier. Customer-level data can be identified in the input dataset by “Customer_ID”, “Account_ID” or a similar item that includes a customer unit or entity identifier. Tradeline-level data is not ordinarily available in raw input datasets and can be determined using a generated contextual item or a combination thereof. For example, referring to FIG. 2D, if a particular contextual item **278** is determined to denote a particular financial institution, the entity extractor **144** can determine the corresponding tradeline subtype **726** (auto loan payment, credit card payment, or business loan payment), and a type of the tradeline for the financial institution (financial tradeline **727**). If a particular contextual item **278** is determined to denote a particular merchant, the entity extractor **144** can determine the corresponding tradeline subtype **726** (telecom/internet payment, bill payment, auto insurance) and a type of the tradeline for the merchant (non-financial tradeline **274**).

As another example, the contextualizer **140** can, at **208b**, perform classification operations using the classifier **146** (for example, as described in reference to FIG. 3A). For example, a classification model can analyze the generated contextual items, such as merchant identifiers, dates, debit or credit transaction type, channels, locations, to determine how the contextual items relate to various transaction class descriptors, such as classifications **224** discussed in relation to FIG. 2B. In some implementations, the classifier **146** can execute operations of the feature definition engine **145**, which can reference a base feature store **145a**, a custom feature store **145b**, and various additional ontologies (e.g.,

based on additional data received from entities such as credit scoring bureaus, asset databases, property databases) to cross-reference a particular contextual item to a particular transaction class descriptor. Example transaction class descriptors can include items such as income/revenue transaction class descriptors, debt payment transaction class descriptors, bill payment transaction class descriptors, spend transaction class descriptors, and/or fee transaction class descriptors.

As another example, the contextualizer **140** can, at **208n**, perform item recurrence identification operations using the recurrence identifier **148**. For example, a set of similar transactions can be identified using fuzzy logic, where the transactions are determined to have similar values for a particular contextual item. Transactions that have similar values for a particular contextual item can be said to belong to the same category (e.g., same merchant, same transaction type—debit or credit, same channel, same location, same tradeline, same transaction class descriptor, and so forth). Then, the recurrence identifier **148** can execute operations to infer recurrence patterns by determining date differences (e.g., in days, months, and so forth) between any two transactions in the set, where the transactions are ordered to occur one after another without intervening transactions of similar type. Accordingly, the recurrence identifier **148** can determine various recurrence related properties, such as recurrence type (1—monthly, 2—weekly, 3—biweekly, 4—daily), recurrence frequency/periodicity (biweekly, monthly, and so forth), autopay y/n, and so forth.

In an example, the recurrence identifier **148** can parse the input data and generate contextual items to obtain a dataset with the following columns: debit/credit, transaction amount, transaction date, description. The transactions in the dataset can be binned (grouped) according, for example, to the following: description, description counter, and amount counter. The description counter and transaction counter can, upon validation, be used to categorize the data into example categories, such as D1A1 (description 1, amount 1), D1A2 (description 1, amount 2), or D1A3 (description 1, amount 3). Then, transactions in each bin can be tagged with a particular recurrence frequency inferred based on the binning range (e.g., the determined periodicity of transactions, such as, for example, if a date difference between two successive transactions is 14 days, it can be inferred that the periodicity is “biweekly”). In some implementations, the periodicity calculation can adjust for weekends.

The resulting dataset can in the following example format:

Description	RecurrenceType	RecurrenceFrequency
ORIG CO NAME: QPAY INC ORIG ID: 14 . . .	1	Bi Weekly
ORIG CO NAME: MERCHANT SERVICE ORIG ID: 18 . . .	1	Monthly
ORIG CO NAME: MERCHANT SERVICE ORIG ID: 18 . . .	2	Monthly
USAA FUNDS TRANSFER CR	3	Bi Weekly

In some implementations, the next set of operations parses the Description column using fuzzy logic or another suitable technique, such as elastic search, rule-based AI techniques, and so forth, to identify a grouping value for the transaction, such as merchant, tradeline, transaction class descriptor, location, and so forth. In some implementations,

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the grouping value for the transaction is identified before the recurrence type and recurrence frequency are determined.

At operations **210**, the summarizer **150** of the data analytics and contextualization platform **110** of FIG. **1A** generates one or more entity behavior maps **154**. As described in relation to FIG. **1A**, the generated entity behavior maps can be transaction-level, tradeline-level, and/or customer-level. Referring to FIG. **1E** as an example (a customer-level entity behavior map), generating a particular entity behavior map at this stage can include generating a tabular or key-value pair dataset that includes column labels (keys). The column labels to be included can be provided by the feature definition engine based on settings for a particular customer (e.g., target application **104a**). Example column labels (keys), as shown, can include a customer identifier, auto loan indicator y/n, and so forth. At this stage the summarizer **150** can populate (add values for) certain items. For example, items that have been parsed from the input data (e.g., customer_id) can be populated. Items that have been determined or generated by contextualization operations (e.g., transaction class description, such as “electricity/gas/power payment count”) can also be populated.

At operations **212**, the summarizer **150** of the data analytics and contextualization platform **110** of FIG. **1A** performs summarization operations. As part of performing summarization operations, a particular entity behavior map can be populated with the corresponding values. Continuing the example, the count of utility payments can be set to a particular value (as shown, 4) by determining the number of corresponding tradelines for the customer identified by the customer_id. More generally, the calculations generated by the summarizer **150** can include any appropriate statistical operations or aggregations, such as cash-in-hand days, credit count, debit count, debt repayment amounts, non-sufficient funds statistics, overdraft limit, and so forth.

At operations **214**, the insight generator **160** of the of the data analytics and contextualization platform **110** of FIG. **1A** performs higher-level insight generation operations. The items of the entity behavior map that are populated at this stage can include various scores/indexes **164**, smart variables **166**, and/or indicators **168** generated by the insight generator **160**, such as commercial loyalty index, commercial risk index, and so forth. The insight generator can perform various AI/ML modeling operations to populate the maps with these items. For example, recurrence analytics can include executing AI/ML operations to identify payment periodicity and determine whether payments have been missed (e.g., not made according to the identified periodicity). Item tradeline analytics and/or descriptor class analytics can include executing AI/ML operations to compute scores or indexes, such as commercial risk indexes (e.g., computed based on factors such as transaction amount, account balance, number of missed payments). Item location analytics can include executing AI/ML operations to generate predictions regarding customer behavior based on determined locations of past transactions. Entity income analytics can include executing AI/ML operations to determine a share or patterns (e.g., to generate projections based on periodicity) of recurring income or revenue for a particular group of transactions (e.g., for transactions associated with a particular category that maps to an extracted token, such as retail sales). Entity employer analytics can include executing AI/ML operations to generate risk scores associated with an employer characteristic. An employer can be determined based on an extracted token. Additional data sources can be referenced to determine company size, years in business, solvency scores, historical stock prices and so forth. Based

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on this data, the platform can generate predictions, such as employee stability scores, income stability scores, and so forth.

At operations **216**, the publisher **170** of the data analytics and contextualization platform **110** of FIG. **1A** publishes one or more output datasets, which can be received or otherwise accessed by an application **104a** of a target computing system **104**. The output dataset(s) can include the elements sufficient for the receiving system to process and/or visualize the corresponding entity behavior map as well as its corresponding indexes, smart variables, indicators, and so forth. For example, the output dataset(s) can be structured to include transaction-level items (as in FIG. **1C**), tradeline-level items (as in FIG. **1D**), or customer-level items (as in FIG. **1E**). The various items can include the corresponding outputs of summarization operations **212** and/or insight generator operations **214**. These outputs can include, for example, calculated, determined, or predicted indexes **164**, smart variables **166**, and/or indicators **168**, such as the items shown in FIGS. **1C-1E**.

In some implementations, the publisher **170** can generate an API message that includes a particular output dataset. The API message can further include metadata. In some implementations, the metadata can include an API type, which can specify whether the API message should be transmitted to the target computing system **104** in real time or included in a batch of API messages for periodic (e.g., hourly, daily) transmission. In some implementations, the metadata can include an output type, which can specify a particular type of entity behavior map on which the API message is based, such as transaction-level, tradeline-level, and/or customer-level. In some implementations, the metadata can include a target application identifier for the application **104a**. The application identifier can include routing information in the form of any of a hyperlink uniquely generated for the target application, an IP address, a port number, or any combination thereof. In some implementations, the metadata can include a set of use case identifiers, which can include reference(s) to particular dimension shown in FIG. **2E**. Accordingly, various output datasets can be customized (e.g., for particular target applications **104a**) to include specific particular dimension data, which optimizes the size of output datasets transmitted via a network.

Example Embodiments of the AI/ML Engines of the Data Analytics and Contextualization Platform

FIG. **3A** illustrates a layered architecture of an artificial intelligence (AI) system **300** that can implement the machine learning models of the data analytics and contextualization platform **110** of FIG. **1A**, in accordance with some implementations of the present technology. FIG. **3B** illustrates an example fuzzy logic engine of the data analytics and contextualization platform **110** of FIG. **1A**, in accordance with some implementations of the present technology. For example, the AI/ML engine(s) **132**, **142**, **144**, **146**, **148**, **152**, and/or **162** can include some or all elements described in relation to FIGS. **3A** and/or **3B**. Additionally, elements of FIG. **3B** can be implemented according to the infrastructure described in relation to FIG. **3A**.

As shown according to FIG. **3A**, the AI system **300** can include a set of layers, which conceptually organize elements within an example network topology for the AI system's architecture to implement a particular AI model. Generally, an AI model is a computer-executable program implemented by the AI system **300** that analyzes data to make predictions. In some implementations, the AI model can include a fuzzy logic engine **340** of FIG. **3B** or another type of logic. In some implementations, the AI model can

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include various other models, such as neural networks trained to identify entities in pre-processed input data, classify entities in pre-processed input data, identify recurrence and other patterns in pre-processed input data, generate indexes, generate smart variables, generate indicators, and so forth.

In the AI model, information can pass through each layer of the AI system 300 to generate outputs for the AI model. The layers can include a data layer 302, a structure layer 304, a model layer 306, and an application layer 308. The algorithm 316 of the structure layer 304 and the model structure 320 and model parameters 322 of the model layer 306 together form an example AI model. The optimizer 326, loss function engine 324, and regularization engine 328 work to refine and optimize the AI model, and the data layer 302 provides resources and support for application of the AI model by the application layer 308.

The data layer 302 acts as the foundation of the AI system 300 by preparing data for the AI model. As shown, the data layer 302 can include two sub-layers: a hardware platform 310 and one or more software libraries 312. The hardware platform 310 can be designed to perform operations for the AI model and can include computing resources for storage, memory, logic and networking, such as the resources described in relation to FIGS. 4 and 5. The hardware platform 310 can process amounts of data using one or more servers. The servers can perform backend operations such as matrix calculations, parallel calculations, machine learning (ML) training, and the like. Examples of servers used by the hardware platform 310 include central processing units (CPUs) and graphics processing units (GPUs). CPUs are electronic circuitry designed to execute instructions for computer programs, such as arithmetic, logic, controlling, and input/output (I/O) operations, and can be implemented on integrated circuit (IC) microprocessors. GPUs are electric circuits that were originally designed for graphics manipulation and output but may be used for AI applications due to their vast computing and memory resources. GPUs use a parallel structure that generally makes their processing more efficient than that of CPUs. In some instances, the hardware platform 310 can include Infrastructure as a Service (IaaS) resources, which are computing resources (e.g., servers, memory, etc.) offered by a cloud services provider. The hardware platform 310 can also include computer memory for storing data about the AI model, application of the AI model, and training data for the AI model. The computer memory can be a form of random-access memory (RAM), such as dynamic RAM, static RAM, and non-volatile RAM.

The software libraries 312 can be thought of suites of data and programming code, including executables, used to control and optimize the computing resources of the hardware platform 310. The programming code can include low-level primitives (e.g., fundamental language elements) that form the foundation of one or more low-level programming languages, such that servers of the hardware platform 310 can use the low-level primitives to carry out specific operations. The low-level programming languages do not require much, if any, abstraction from a computing resource's instruction set architecture, allowing them to run quickly with a small memory footprint. Examples of software libraries 312 that can be included in the AI system 300 include Intel Math Kernel Library, Nvidia cuDNN, Eigen, and Open BLAS. In some implementations, a software library 312 can include executables to optimize performance of various fuzzy logic operations performed by the fuzzy logic engine 340 of FIG. 3B.

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The structure layer 304 can include an ML framework 314 and an algorithm 316. The ML framework 314 can be thought of as an interface, library, or tool that allows users to build and deploy the AI model. The ML framework 314 can include an open-source library, an application programming interface (API), a gradient-boosting library, an ensemble method, and/or a deep learning toolkit that work with the layers of the AI system facilitate development of the AI model. For example, the ML framework 314 can distribute processes for application or training of the AI model across multiple resources in the hardware platform 310. The ML framework 314 can also include a set of pre-built components that have the functionality to implement and train the AI model and allow users to use pre-built functions and classes to construct and train the AI model. Thus, the ML framework 314 can be used to facilitate data engineering, development, hyperparameter tuning, testing, and training for the AI model. Examples of ML frameworks 314 that can be used in the AI system 300 include TensorFlow, PyTorch, Scikit-Learn, Scikit-Fuzzy, Keras, Caffe, LightGBM, Random Forest, Fuzzy Logic Toolbox, and Amazon Web Services.

The algorithm 316 can be an organized set of computer-executable operations used to generate output data from a set of input data and can be described using pseudocode. The algorithm 316 can include program code that allows the computing resources to learn from new input data and create new/modified outputs based on what was learned. In some implementations, the algorithm 316 can build the AI model through being trained while running computing resources of the hardware platform 310. This training allows the algorithm 316 to make predictions or decisions without being explicitly programmed to do so. Once trained, the algorithm 316 can run at the computing resources as part of the AI model to make predictions or decisions, improve computing resource performance, or perform tasks. The algorithm 316 can be trained using supervised learning, unsupervised learning, semi-supervised learning, and/or reinforcement learning.

Using supervised learning, the algorithm 316 can be trained to learn patterns (e.g., match input data to output data) based on labeled training data, such as transaction categorization data, entity behavior map data, and so forth.

Supervised learning can involve classification and/or regression. Classification techniques involve teaching the algorithm 316 to identify a category of new observations based on training data and are used when the input data for the algorithm 316 is discrete. Said differently, when learning through classification techniques, the algorithm 316 receives training data labeled with categories (e.g., classes) and determines how features observed in the training data (e.g., merchant identifiers, dates, debit or credit transaction type, channels, locations, and so forth) relate to the categories, such as classifications 224 discussed in relation to FIG. 2B. Once trained, the algorithm 316 can categorize new data by analyzing the new data for features that map to the categories. Examples of classification techniques include boosting, decision tree learning, genetic programming, learning vector quantization, k-nearest neighbor (k-NN) algorithm, and statistical classification.

Regression techniques involve estimating relationships between independent and dependent variables and are used when input data to the algorithm 316 is continuous. Regression techniques can be used to train the algorithm 316 to predict or forecast relationships between variables. To train the algorithm 316 using regression techniques, a user can select a regression method for estimating the parameters of

the model. The user collects and labels training data that is input to the algorithm **316** such that the algorithm **316** is trained to understand the relationship between data features and the dependent variable(s). Once trained, the algorithm **316** can predict missing historic data or future outcomes based on input data. Examples of regression methods include linear regression, multiple linear regression, logistic regression, regression tree analysis, least squares method, and gradient descent. In an example implementation, regression techniques can be used, for example, to estimate and fill-in missing data for machine-learning based pre-processing operations. In another example implementation, regression techniques can be used to generate predictions for various recurrence items **226** of FIG. 2B or other similar items.

Under unsupervised learning, the algorithm **316** learns patterns from unlabeled training data. In particular, the algorithm **316** is trained to learn hidden patterns and insights of input data, which can be used for data exploration or for generating new data. Here, the algorithm **316** does not have a predefined output, unlike the labels output when the algorithm **316** is trained using supervised learning. Said another way, unsupervised learning is used to train the algorithm **316** to find an underlying structure of a set of data, group the data according to similarities, and represent that set of data in a compressed format. The platform can use unsupervised learning to identify patterns in input data, such as the patterns relating to recurrence items **226** of FIG. 2B.

The model layer **306** implements the AI model using data from the data layer and the algorithm **316** and ML framework **314** from the structure layer **304**, thus enabling decision-making capabilities of the AI system **300**. The model layer **306** can include a model structure **320**, model parameters **322**, a loss function engine **324**, an optimizer **326**, and/or a regularization engine **328**.

The model structure **320** describes the architecture of the AI model of the AI system **300**. The model structure **320** defines the complexity of the pattern/relationship that the AI model expresses. Examples of structures that can be used as the model structure **320** include decision trees, support vector machines, regression analyses, Bayesian networks, Gaussian processes, genetic algorithms, and artificial neural networks (or, simply, neural networks). The model structure **320** can include a number of structure layers, a number of nodes (or neurons) at each structure layer, and activation functions of each node. Each node's activation function defines how to node converts data received to data output. The structure layers may include an input layer of nodes that receive input data, an output layer of nodes that produce output data. The model structure **320** may include one or more hidden layers of nodes between the input and output layers. The model structure **320** can be an Artificial Neural Network (or, simply, neural network) that connects the nodes in the structured layers such that the nodes are interconnected. Examples of neural networks include Feed-forward Neural Networks, convolutional neural networks (CNNs), Recurrent Neural Networks (RNNs), Autoencoder, and Generative Adversarial Networks (GANs). According to various implementations, neural networks can be used, for example, by the insight generator **160** of FIG. 1A.

The model parameters **322** represent the relationships learned during training and can be used to make predictions and decisions based on input data. The model parameters **322** can weight and bias the nodes and connections of the model structure **320**. For instance, when the model structure **320** is a neural network, the model parameters **322** can weight and bias the nodes in each layer of the neural

networks, such that the weights determine the strength of the nodes and the biases determine the thresholds for the activation functions of each node. The model parameters **322**, in conjunction with the activation functions of the nodes, determine how input data is transformed into desired outputs. The model parameters **322** can be determined and/or altered during training of the algorithm **316**.

The loss function engine **324** can determine a loss function, which is a metric used to evaluate the AI model's performance during training. For instance, the loss function engine **324** can measure the difference between a predicted output of the AI model and the actual output of the AI model and is used to guide optimization of the AI model during training to minimize the loss function.

The optimizer **326** adjusts the model parameters **322** to minimize the loss function during training of the algorithm **316**. In other words, the optimizer **326** uses the loss function generated by the loss function engine **324** as a guide to determine what model parameters lead to the most accurate AI model. Examples of optimizers include Gradient Descent (GD), Adaptive Gradient Algorithm (AdaGrad), Adaptive Moment Estimation (Adam), Root Mean Square Propagation (RMSprop), Radial Base Function (RBF) and Limited-memory BFGS (L-BFGS). The type of optimizer **326** used may be determined based on the type of model structure **320** and the size of data and the computing resources available in the data layer **302**.

The regularization engine **328** executes regularization operations. Regularization is a technique that prevents over- and under-fitting of the AI model. Overfitting occurs when the algorithm **316** is overly complex and too adapted to the training data, which can result in poor performance of the AI model. Underfitting occurs when the algorithm **316** is unable to recognize even basic patterns from the training data such that it cannot perform well on training data or on validation data. The optimizer **326** can apply one or more regularization techniques to fit the algorithm **316** to the training data properly, which helps constraint the resulting AI model and improves its ability for generalized application. Examples of regularization techniques include lasso (L1) regularization, ridge (L2) regularization, and elastic (L1 and L2 regularization).

The application layer **308** describes how the AI system **300** is used to solve problem or perform tasks. In an example implementation, the application layer **308** can include the publisher **170** of the data analytics and contextualization platform **110** of FIG. 1A. For example, the publisher **170** can generate data files and/or messages that include model outputs.

FIG. 3B illustrates an example fuzzy logic engine **340** of the data analytics and contextualization platform **110** of FIG. 1A, in accordance with some implementations of the present technology. According to various implementations, the AI/ML engine(s) **132**, **142**, **144**, **146**, **148**, **152**, and/or **162** can include elements of the fuzzy logic engine **340**. For example, the AI/ML engine **142** of the contextualizer **140** can use fuzzy logic on unstructured data items extracted from input data in order to generate a set of contextual items. Contextual items can include, for example, various items described in relation to FIG. 2B, such as classification items **224**, recurrence items **226**, entity/merchant **228**, channel **230**, location **240**, and so forth.

The fuzzy logic engine **340** can be structured to perform textual analysis using natural language processing (NLP) or a similar technique. The textual analysis can include standardizing various items (tokens) extracted from input data, such as transaction records (**192a**, **192b**, **192c**) of the input

dataset **192** of FIG. 1B, raw transaction data **222** of FIG. 2B, transaction descriptions **252** of FIG. 2C, and so forth. For example, the fuzzy logic engine **340** can extract a set of abbreviated items **252a** from transaction descriptions **252** for at least two transactions of a set of transactions. In some implementations, if the extracted abbreviated items **252a** meet a predetermined threshold of similarity (e.g., a minimum count of similar consecutive characters, a minimum count of similar consecutive phrases, an overlap by at least a threshold amount of positions within transaction descriptions **252** and so forth), then the extracted abbreviated items **252a** are determined to match (i.e. can both/all be identified as a particular contextual item, such as a merchant entity descriptor). In some implementations, if the extracted abbreviated items **252a** correspond to a particular item in an ontology, such as an ontology of a base feature store **145a** and/or a custom feature store **145b**, then the extracted abbreviated items **252a** are also determined to match. For example, a first extracted abbreviated item **252a** can include a character sequence “AMZN” and a second extracted abbreviated item **252a** can include a character sequence “AMAZON”. If both sequences can be found in a particular ontology and correspond to the same entity (e.g., “Amazon.com”), then an inference can be made that the items match. These items can then be classified (**258**, **260**) and/or recurrence patterns (**262**, **264**, **280**, **282**, **284**) can be determined therefor.

To carry out the aforementioned operations, the fuzzy logic engine **340** can include a fuzzifier **346**, which accepts input items **344** (e.g., tokens/items extracted from raw transaction data **222** of FIG. 2B, transaction description **252** of FIG. 2C, and so forth). The input items can include various attributes, such as, for example, item length, item location, presence of a particular character sequence, and so forth. For example, a particular input item can include the character sequence “AMZN” in the 0-3 position. The fuzzifier **346** can apply a membership function **347** to a particular attribute or combination of attributes to classify the particular attribute of combination of attributes. For example, characters in the 0-3 position can be matched to one of a set of numerical values in the range 0-1 that denote the likelihood of the characters denoting a merchant according to the rules **348**. The rules **348**, for example, can be based on various data interchange formats, such as the Open Banking, ACH, and so forth, which can specify locations, allowable values, and so forth for various items in transaction data such that items can be identified based on their respective locations within input data (e.g., “if an item is in the 0-3 position, the item denotes a merchant”). As another example, the character sequence “AMZN” can denote a merchant even if not found in a particular position based on, for example, the determined similarity of the character sequence to an item in a reference ontology. The determined similarity can be based on a degree of certainty, such as a minimum count of consecutive characters corresponding to a particular degree of certainty, and a higher-than-minimum count corresponding to a comparatively higher degree of certainty.

More generally, the inference engine **350** can use the rules **348** on the output of the fuzzifier **346** in order to determine the degree of correspondence between the input item **344** and the rules **348**. The defuzzifier **352** can generate an output item **354** based on the rules **348**. Continuing the example, the rules **348** can state that if a particular item is found in a particular position, or if a particular character sequence in any position is found in a reference ontology that lists merchant entities, then the input item **344** relates to a

particular merchant entity. The output item **354** can, for example, output an ontology item and its corresponding value (e.g., “Amazon.com.”)

One of skill will appreciate that the inference engine **350** can be configured to access appropriate rule sets to various other contextual items, such as classification items **224**, recurrence items **226** (amount, time, type, frequency), entity/merchant **228**, channel **230**, and/or location **240** of FIG. 2B, or other items.

Example Computing Environment of the Data Analytics and Contextualization Platform

FIG. 4 is a block diagram showing some of the components typically incorporated in at least some of the computer systems and other devices **400** on which the disclosed data analytics and contextualization platform operates in accordance with some implementations of the present technology. As shown, an example computer system **400** can include: one or more processors **402**, main memory **408**, non-volatile memory **410**, a network interface device **414**, video display device **420**, an input/output device **422**, a control device **424** (e.g., keyboard and pointing device), a drive unit **426** that includes a machine-readable medium **428**, and a signal generation device **432** that are communicatively connected to a bus **418**. The bus **418** represents one or more physical buses and/or point-to-point connections that are connected by appropriate bridges, adapters, or controllers. Various common components (e.g., cache memory) are omitted from FIG. 4 for brevity. Instead, the computer system **400** is intended to illustrate a hardware device on which components illustrated or described relative to the examples of the figures and any other components described in this specification can be implemented.

The computer system **400** can take any suitable physical form. For example, the computer system **400** can share a similar architecture to that of a server computer, personal computer (PC), tablet computer, mobile telephone, game console, music player, wearable electronic device, network-connected (“smart”) device (e.g., a television or home assistant device), AR/VR systems (e.g., head-mounted display), or any electronic device capable of executing a set of instructions that specify action(s) to be taken by the computer system **400**. In some implementations, the computer system **400** can be an embedded computer system, a system-on-chip (SOC), a single-board computer system (SBC) or a distributed system such as a mesh of computer systems or include one or more cloud components in one or more networks. Where appropriate, one or more computer systems **400** can perform operations in real-time, near real-time, or in batch mode.

The network interface device **414** enables the computer system **400** to exchange data in a network **416** with an entity that is external to the computing system **400** through any communication protocol supported by the computer system **400** and the external entity. Examples of the network interface device **414** include a network adaptor card, a wireless network interface card, a router, an access point, a wireless router, a switch, a multilayer switch, a protocol converter, a gateway, a bridge, bridge router, a hub, a digital media receiver, and/or a repeater, as well as all wireless elements noted herein.

The memory (e.g., main memory **408**, non-volatile memory **412**, machine-readable medium **428**) can be local, remote, or distributed. Although shown as a single medium, the machine-readable medium **428** can include multiple media (e.g., a centralized/distributed database and/or associated caches and servers) that store one or more sets of instructions **430**. The machine-readable (storage) medium

428 can include any medium that is capable of storing, encoding, or carrying a set of instructions for execution by the computer system 400. The machine-readable medium 428 can be non-transitory or comprise a non-transitory device. In this context, a non-transitory storage medium can include a device that is tangible, meaning that the device has a concrete physical form, although the device can change its physical state. Thus, for example, non-transitory refers to a device remaining tangible despite this change in state.

Although implementations have been described in the context of fully functioning computing devices, the various examples are capable of being distributed as a program product in a variety of forms. Examples of machine-readable storage media, machine-readable media, or computer-readable media include recordable-type media such as volatile and non-volatile memory, removable memory, hard disk drives, optical disks, and transmission-type media such as digital and analog communication links.

In general, the routines executed to implement examples herein can be implemented as part of an operating system or a specific application, component, program, object, module, or sequence of instructions (collectively referred to as “computer programs”). The computer programs typically comprise one or more instructions (e.g., instructions 410, 430) set at various times in various memory and storage devices in computing device(s). When read and executed by the processor 402, the instruction(s) cause the computer system 400 to perform operations to execute elements involving the various aspects of the disclosure.

FIG. 5 is a system diagram illustrating an example of a computing environment in which the disclosed data analytics and contextualization platform operates in some implementations. In some implementations, environment 500 includes one or more client computing devices 505A-D, examples of which can host the data analytics and contextualization platform 110 of FIG. 1A. Client computing devices 505 operate in a networked environment using logical connections through network 530 to one or more remote computers, such as a server computing device.

In some implementations, server 510 is an edge server which receives client requests and coordinates fulfillment of those requests through other servers, such as servers 520A-C. In some implementations, server computing devices 510 and 520 comprise computing systems, such as the data analytics and contextualization platform 110 of FIG. 1A. Though each server computing device 510 and 520 is displayed logically as a single server, server computing devices can each be a distributed computing environment encompassing multiple computing devices located at the same or at geographically disparate physical locations. In some implementations, each server 520 corresponds to a group of servers.

Client computing devices 505 and server computing devices 510 and 520 can each act as a server or client to other server or client devices. In some implementations, servers (510, 520A-C) connect to a corresponding database (515, 525A-C). As discussed above, each server 520 can correspond to a group of servers, and each of these servers can share a database or can have its own database. Databases 515 and 525 warehouse (e.g., store) information such as transaction data, feature data, entity data, classifier data, recurrence data, entity behavior map data, indexes, smart variables, indicators, configuration data and so forth. Though databases 515 and 525 are displayed logically as single units, databases 515 and 525 can each be a distributed computing environment encompassing multiple computing

devices, can be located within their corresponding server, or can be located at the same or at geographically disparate physical locations.

Network 530 can be a local area network (LAN) or a wide area network (WAN), but can also be other wired or wireless networks. In some implementations, network 530 is the Internet or some other public or private network. Client computing devices 505 are connected to network 530 through a network interface, such as by wired or wireless communication. While the connections between server 510 and servers 520 are shown as separate connections, these connections can be any kind of local, wide area, wired, or wireless network, including network 530 or a separate public or private network.

Example Use Cases of the Data Analytics and Contextualization Platform

Various use cases for the data analytics and contextualization platform 110 of FIG. 1A are described in relation to FIGS. 6-10.

According to an example use case 600 of FIG. 6, inferences 610 can be made about a particular customer based on input data items. Input data items, included in one or more transaction datasets, such as credit card transactions, debit card transactions, deposit account transactions, and so forth, can include a date 612, amount 614, description 616, and/or transaction type 618. The platform can parse the description fields to extract various tokens. The tokens can, for example, correspond to various transaction entity/merchant identifiers (e.g., “SHOPIFY”, “AMAZON”, “CHASE CREDIT”, “VZ WIRELESS”). Based on the entity identifier, which can be extracted using fuzzy logic and cross-referenced to an ontology, various transaction categories can be determined. As shown, the categories can include revenue 620 and tradelines (financial products and utility) 630. For example, transactions of a particular type (debit/credit) can correspond to employer or other revenue payments (in this use case, revenue from retail sales). The revenue periodicity 624 can be determined using the transaction date in combination with transaction type and/or with a cross-referenced token extracted from transaction description. A share of recurring income or revenue 622 can be determined for a particular group of transactions (e.g., for transactions associated with a particular category that maps to an extracted token, such as retail sales). Further, as shown, the categories can include various tradelines 630, and the platform can use the transaction amounts 614 to calculate payment percentages 632 (e.g., a ratio of a payment to account balance, a ratio of a payment to account revenue for the corresponding period). The platform can also use payment periodicity values to determine whether any payments due were missed in a particular time period 634. The platform can also cross-reference additional data (e.g., credit data, Open Banking data) to determine tradeline balances and terms 630.

According to an example use case 700 of FIG. 7, inferences 702-712 can be made about a particular customer based on input data items. Input data items, included in one or more transaction datasets, such as credit card transactions, debit card transactions, deposit account transactions, and so forth, can include a description 720, amount 722, datetime 724 (including, for example, a date 732), and transaction type 726. The platform can parse the description fields to extract various token, such as entity description 730, which can be extracted using fuzzy logic and cross-referenced to an ontology. Here, at 702, entity extraction can identify a particular token. At 704, using the token (e.g., using fuzzy logic to map the token to a particular item in an ontology), it can be determined that the token describes a particular

category (e.g., collection agency). At **706**, other contextual items, such as channel (online), periodicity (monthly) and amount (\$445) can be determined to describe tradeline-level activity. At **708**, recurrence analysis based on periodicity can make an inference that a particular payment pattern (a gap of two months for monthly payments) indicates a missed monthly payment. At **710**, customer-level indexes and values, such as percentage **728** of gross monthly revenue **730**, can be generated/determined. At **712**, additional customer-level calculations, such as cash-in-hand days **732**, monthly revenue projections **734** (e.g., gross monthly revenue minus projected cash outflows), and so forth, can be performed.

According to an example use case **800** of FIG. **8**, inferences **810** can be made about a particular customer based on input data items. Input data items, included in one or more transaction datasets, such as credit card transactions, debit card transactions, deposit account transactions, and so forth, can include a date **812**, amount **814**, description **816**, and/or transaction type **818**. The platform can parse the description fields to extract various tokens. The tokens can, for example, correspond to various transaction entity/merchant identifiers (e.g., "PIZZA", "SERVICE STATION"). Based on the entity identifier, which can be extracted using fuzzy logic and cross-referenced to an ontology, various transaction categories can be determined. As shown, the categories can include revenue **820**, tradelines (financial products and utility) **830**, and consumer spending (e.g., discretionary spending) **840**. The amounts **814** can be used to calculate shares **842** of spending by spending category. Spending categories can be determined by cross-referencing tokens to ontologies to identify merchants and determine merchant categories.

According to an example use case **900** of FIG. **9**, inferences **910-914** can be made about a particular tradeline or customer based on input data items. Input data items, included in one or more transaction datasets, such as credit card transactions, debit card transactions, deposit account transactions, and so forth, can include a date, amount, description, and/or transaction type. After determining a particular tradeline using input data items (as described, for example, in reference to FIG. **6** and elsewhere herein), the platform can cross-reference (e.g., using a customer identifier corresponding to a transaction or set of transactions that served as the basis for determining the tradeline) third-party data, such as credit data, account data, and so forth, to determine overall outstanding balance **902**. Here, the outstanding balance **902** is a loan balance for a lender. The platform can perform analytics on the input data to cross-reference the outstanding balance to payment amounts at various points in time in order to generate an outstanding balance trendline **906**. The platform can further cross-reference various data points of interest on the trendline (for example, by referencing account data, such as when a particular account was open, acquired by a particular entity, became delinquent, and so forth) to visualize the relationship of these data points to the trendline **906**.

According to an example use case **1000** of FIG. **10**, input transaction data, such as description **1002**, can be tokenized using fuzzy logic and cross-referenced to an ontology in order to determine and label the transaction with various contextual items, such as merchant/institution **1006**, city **1008**, state **1010**, and category **1012**. The contextual items, such as transaction category, **1012**, can be used to make inferences about demographics **1020**, financial health **1030**, personality **1040**, payment behavior **1050** and so forth for a particular set of transactions.

CONCLUSION

Unless the context clearly requires otherwise, throughout the description and the claims, the words "comprise," "com-

prising," and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of "including, but not limited to." As used herein, the terms "connected," "coupled," or any variant thereof means any connection or coupling, either direct or indirect, between two or more elements; the coupling or connection between the elements can be physical, logical, or a combination thereof. Additionally, the words "herein," "above," "below," and words of similar import, when used in this application, refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively. The word "or," in reference to a list of two or more items, covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list.

The above Detailed Description of examples of the technology is not intended to be exhaustive or to limit the technology to the precise form disclosed above. While specific examples for the technology are described above for illustrative purposes, various equivalent modifications are possible within the scope of the technology, as those skilled in the relevant art will recognize. For example, while processes or blocks are presented in a given order, alternative embodiments may perform routines having steps, or employ systems having blocks, in a different order, and some processes or blocks may be deleted, moved, added, subdivided, combined, and/or modified to provide alternative or sub-combinations. Each of these processes or blocks may be implemented in a variety of different ways. Also, while processes or blocks are at times shown as being performed in series, these processes or blocks may instead be performed or implemented in parallel, or may be performed at different times. Further, any specific numbers noted herein are only examples: alternative embodiments may employ differing values or ranges.

The teachings of the technology provided herein can be applied to other systems, not necessarily the system described above. The elements and acts of the various examples described above can be combined to provide further embodiments of the technology. Some alternative embodiments of the technology may include not only additional elements to those embodiments noted above, but also may include fewer elements.

These and other changes can be made to the technology in light of the above Detailed Description. While the above description describes certain examples of the technology, and describes the best mode contemplated, no matter how detailed the above appears in text, the technology can be practiced in many ways. Details of the system may vary considerably in its specific implementation, while still being encompassed by the technology disclosed herein. As noted above, specific terminology used when describing certain features or aspects of the technology should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features, or aspects of the technology with which that terminology is associated. In general, the terms used in the following claims should not be construed to limit the technology to the specific examples disclosed in the specification, unless the above Detailed Description section explicitly defines such terms. Accordingly, the actual scope of the technology encompasses not only the disclosed examples, but also all equivalent ways of practicing or implementing the technology under the claims.

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To reduce the number of claims, certain aspects of the technology are presented below in certain claim forms, but the applicant contemplates the various aspects of the technology in any number of claim forms. For example, while only one aspect of the technology is recited as a computer-readable medium claim, other aspects may likewise be embodied as a computer-readable medium claim, or in other forms, such as being embodied in a means-plus-function claim. Any claims intended to be treated under 35 U.S.C. § 112 (f) will begin with the words “means for,” but use of the term “for” in any other context is not intended to invoke treatment under 35 U.S.C. § 112 (f). Accordingly, the applicant reserves the right to pursue additional claims after filing this application to pursue such additional claim forms, in either this application or in a continuing application.

We claim:

1. A computing system comprising at least one processor, at least one memory, and one or more non-transitory computer readable media storing instructions, which when executed by the at least one processor, perform operations for artificial intelligence/machine learning (AI/ML) based modeling and simulation and generating a subscriber-customized reduced-size feature set, the operations comprising:

acquiring, by a platform instance of the computing system, an input dataset comprising a set of structured items and unstructured textual data that relates to one or more items in the input dataset; and

reducing a size of the input dataset for transmission, via an electronic communication channel, to a target computing system by generating a labeled enriched feature set that corresponds to a set of automatically determined contextual items, comprising:

extracting the unstructured textual data from the input dataset;

using fuzzy logic, generating, by the platform instance, the set of contextual items using the extracted unstructured textual data;

labeling a set of items in the input dataset with the set of contextual items generated for a particular item in the set of items;

using a labeled particular item from the set of items in the labeled enriched feature set,

generating a set of entity behavior maps comprising one or more of:

a transaction map, a customer map, or a tradeline map,

wherein each entity behavior map in the set of entity behavior maps includes a set of features selected from the labeled enriched feature set based on a first particular label of a particular item in the set of items such that a particular entity behavior map corresponds to a particular contextual item;

for each of the set of entity behavior maps,

using a classification of the entity behavior map, the classification corresponding to the transaction map, the customer map, or the tradeline map, determining a set of operations to populate the entity behavior map, a particular operation in the set of operations being one of (i) AI/ML modeling operations performed by a trained neural network, (ii) an aggregation operation, or (iii) a summarization operation; and

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using the particular contextual item that corresponds to the classification of the entity behavior map, performing operations comprising:

populating the entity behavior map;

using the classification and a target system identifier for the target computing system, querying a feature store to determine a particular dimension of the entity behavior map to transmit to the target computing system, wherein the particular dimension is a portion of the entity behavior map;

generating an electronic message for a reduced-size transmission, wherein the electronic message comprises the particular dimension; and

causing the target computing system to process the reduced-size transmission for publication of the dimension of the entity behavior map at the target computing system.

2. The system of claim 1, wherein generating by the platform instance the set of contextual items comprises:

extracting a token from the unstructured textual data; and referencing an ontology to determine whether the token matches an item in the ontology within a predetermined degree of certainty,

wherein the set of contextual items includes an item descriptor class from the ontology.

3. The system of claim 2,

wherein the ontology corresponds to a feature definition criterion from the feature store associated with the target computing system, the target computing system being a particular subscriber computing system having a particular associated feature store; and

wherein a particular AI/ML modeling operation for the entity behavior map is performed based on the feature definition criterion.

4. The system of claim 3,

wherein performing the AI/ML modeling operation comprises executing a machine learning model, and wherein the machine learning model operates on items extracted from the input dataset,

the operations further comprising:

using the labeled particular item, generating an input feature according to the feature definition criterion corresponding to a particular label; and

using the input feature, executing a machine learning operation to perform the AI/ML modeling operation.

5. The system of claim 4, the AI/ML modeling operation comprising at least one of item recurrence analytics, item tradeline analytics, item descriptor class analytics, item location analytics, entity income analytics, or entity employer analytics.

6. The system of claim 1, the operations comprising acquiring the input dataset from the target computing system.

7. The system of claim 1, the operations comprising acquiring the input dataset from a third-party computing system relative to the target computing system, wherein the input dataset includes data contributed by an entity associated with the target computing system.

8. The system of claim 1, wherein the target computing system is one of a plurality of target computing systems and the platform instance is one of a plurality of platform instances, and wherein each particular target computing system of the plurality of target computing systems is associated with a particular platform instance configured for a particular subscriber.

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9. The system of claim 1, wherein labeling a particular item comprises generating metadata for the particular item using the contextual item.

10. The system of claim 1, the operations comprising:
reducing a size of at least a portion of the input dataset by
quantizing values in the at least a portion of the input
dataset; and
using the quantized values to generate numerical values
for the particular dimension transmitted to the target
computing system.

11. A computer-implemented method for artificial intelligence/machine learning (AI/ML) based modeling and simulation and generating a subscriber-customized reduced-size feature set, the method comprising:

acquiring, by a platform instance of a computing system,
an input dataset comprising a set of structured items
and unstructured textual data that relates to one or more
items in the input dataset;

reducing a size of the input dataset for transmission, via
an electronic communication channel, to a target computing
system by generating a labeled enriched feature
set that corresponds to a set of automatically determined
contextual items, comprising:

extracting the unstructured textual data from the input
dataset; and

using fuzzy logic, generating, by the platform instance,
the set of contextual items using the extracted unstructured
textual data;

labeling a set of items in the input dataset with the set of
contextual items;

using a labeled particular item from the set of items in the
labeled enriched feature set,

generating a set of entity behavior maps comprising one
or more of: a transaction map, a customer map, or a
tradeline map

wherein each entity behavior map in the set of entity
behavior maps includes a set of features selected from
the labeled enriched feature set based on a first particular
label of the particular item such that a particular
entity behavior map corresponds to a particular contextual
item;

for each of the set of entity behavior maps,

using a classification of the entity behavior map, the
classification corresponding to the transaction map, the
customer map, or the tradeline map, determining a set
of operations to populate the entity behavior map, a
particular operation in the set of operations being one
of (i) AI/ML modeling operations performed by a
trained neural network, (ii) an aggregation operation, or
(iii) a summarization operation; and

using the particular contextual item that corresponds to
the classification of the entity behavior map, performing
operations comprising:

populating the entity behavior map;

using the classification and a target system identifier for
the target computing system, querying a feature store
to determine a particular dimension of the entity
behavior map to transmit to the target computing
system, wherein the particular dimension is a portion
of the entity behavior map;

generating an electronic message for a reduced-size
transmission, wherein the electronic message comprises
the particular dimension; and

causing the target computing system to process the
reduced-size transmission for publication of the
dimension of the entity behavior map at the target
computing system.

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12. The method of claim 11, wherein generating by the
platform instance the set of contextual items comprises:

extracting a token from the unstructured textual data; and
referencing an ontology to determine whether the token
matches an item in the ontology within a predetermined
degree of certainty,

wherein the set of contextual items includes an item
descriptor class from the ontology.

13. The method of claim 12,

wherein the ontology corresponds to a feature definition
criterion from the feature store associated with the
target computing system, the target computing system
being a particular subscriber computing system having
a particular associated feature store; and

wherein a particular AI/ML modeling operation for the
entity behavior map is performed based on the feature
definition criterion.

14. The method of claim 13,

wherein performing the AI/ML modeling operation comprises
executing a machine learning model, and
wherein the machine learning model operates on items
extracted from the input dataset,

the operations further comprising:

using the labeled particular item, generating an input
feature according to the feature definition criterion
corresponding to a particular label; and

using the input feature, executing a machine learning
operation to perform the AI/ML modeling operation.

15. The method of claim 14, the AI/ML modeling operation
comprising at least one of item recurrence analytics,
item tradeline analytics, item descriptor class analytics, item
location analytics, entity income analytics, or entity
employer analytics.

16. Method of claim 11, further comprising acquiring the
input dataset from the target computing system.

17. The method of claim 11, further comprising acquiring
the input dataset from a third-party computing system relative
to the target computing system, wherein the input
dataset includes data contributed by an entity associated
with the target computing system.

18. The method of claim 11, wherein the target computing
system is one of a plurality of target computing systems and
the platform instance is one of a plurality of platform
instances, and wherein each particular target computing
system of the plurality of target computing systems is
associated with a particular platform instance configured for
a particular subscriber.

19. The method of claim 11, wherein labeling a particular
item comprises generating metadata for the particular item
using the contextual item.

20. One or more non-transitory computer-readable media
having computer-executable instructions stored thereon, the
instructions, when executed by at least one process of a
computing system, performing operations for artificial intelligence/machine learning (AI/ML) based modeling and simulation and generating a subscriber-customized reduced-size feature set, the operations comprising:

acquiring, by a platform instance of the computing system,
an input dataset comprising a set of structured
items and unstructured textual data that relates to one or
more items in the input dataset; and

reducing a size of the input dataset for transmission, via
an electronic communication channel, to a target computing
system by generating a labeled enriched feature
set that corresponds to a set of automatically determined
contextual items, comprising:

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extracting the unstructured textual data from the input dataset;

using fuzzy logic, generating, by the platform instance, the set of contextual items using the extracted unstructured textual data; 5

labeling a set of items in the input dataset with the set of contextual items generated for a particular item in the set of items;

using a labeled particular item from the set of items in the labeled enriched feature set, 10

generating a set of entity behavior maps comprising one or more of:

a transaction map, a customer map, or a tradeline map 15

wherein each entity behavior map in the set of entity behavior maps includes a set of features selected from the labeled enriched feature set based on a first particular label of a particular item in the set of items such that a particular entity behavior map 20 corresponds to a particular contextual item;

for each of the set of entity behavior maps,

using a classification of the entity behavior map, the classification corresponding to the transaction map, the customer map, or the tradeline map, determining 25 a set of operations to populate the entity behavior map, a particular operation in the set of operations being one of (i) AI/ML modeling operations per-

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formed by a trained neural network, (ii) an aggregation operation, or (iii) a summarization operation; and

using the particular contextual item that corresponds to the classification of the entity behavior map, performing operations comprising:

populating the entity behavior map;

using the classification and a target system identifier for the target computing system, querying a feature store to determine a particular dimension of the entity behavior map to transmit to the target computing system, wherein the particular dimension is a portion of the entity behavior map;

generating an electronic message for a reduced-size transmission, wherein the electronic message comprises the particular dimension; and

causing the target computing system to process a reduced-side transmission for publication of the dimension of the entity behavior map at the target computing system.

21. The media of claim 20, wherein generating by the platform instance the set of contextual items comprises: extracting a token from the unstructured textual data; and referencing an ontology to determine whether the token matches an item in the ontology within a predetermined degree of certainty, wherein the set of contextual items includes an item descriptor class from the ontology.

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